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TECHNICAL SURVEY REPORT

IS ARTIFICIAL INTELLIGENCE DISRUPTING THE WORK–LIFE BALANCE OF HIGHER EDUCATION STUDENTS (AGE 18–30)?

An Empirical Mixed-Methods Investigation into Academic Acceleration, Boundary Permeability, and Technostress

SUBMITTED TO

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ABSTRACT

Background: Artificial Intelligence (AI), particularly generative conversational agents and automated synthesis tools, has rapidly diffused across higher education. While promoting task efficiency, AI fundamentally alters academic rhythms, study schedules, and student well-being.

Objective: This study investigates whether AI-enabled tools disrupt or support the work–life balance (WLB) of higher education students aged 18–30.

Methods: An empirical cross-sectional survey instrument comprising 22 closed- and open-ended questions was administered via Google Forms to 40 validated student respondents. Quantitative data were analyzed through frequency distributions, while qualitative responses underwent 3-level conceptual and relational content analysis.

Results: While 62.5% of respondents report reduced study hours and 85.0% note superior output quality, boundary erosion is pervasive. Nearly three-quarters (72.5%) experience moderate-to-large interference with personal and family life, 45.0% regularly work outside normal hours, and 72.5% suffer physical discomfort. Moreover, only 45.0% confirm clear institutional AI policies, whereas 87.5% assert that ethical transparency would bolster trust.

Conclusion: AI is a double-edged sword: it accelerates task completion but expands workload expectations and dissolves temporal boundaries. Sustainable student balance requires clear institutional right-to-disconnect guidelines and ethical AI literacy.

Keywords: *Artificial Intelligence; Work–Life Balance; Higher Education; Technostress; Boundary Permeability; Academic Burnout; Mixed-Methods Survey.*

Is Artificial Intelligence Disrupting the Work–Life Balance of Higher Education Students (Age 18–30)?

1. INTRODUCTION

1.1 Background and Context

The contemporary higher education landscape is undergoing a profound paradigm shift driven by the ubiquitous integration of Artificial Intelligence (AI). Technologies ranging from large language models (LLMs) such as OpenAI's ChatGPT, Anthropic's Claude, and Microsoft Copilot to automated code synthesizers, intelligent bibliographic managers, and algorithmic scheduling assistants have transitioned from novel curiosities to standard academic equipment. Higher education students, particularly those enrolled in intensive undergraduate, postgraduate, and doctoral curricula, routinely interact with AI tools to conceptualize complex academic problems, debug software code, summarize dense academic literature, draft project reports, and translate multilingual texts. For students aged 18 to 30, this technological revolution coincides with a critical developmental and career transition. Individuals within this demographic cohort are not only expected to fulfill demanding coursework, laboratory assignments, capstone projects, and thesis examinations, but are also increasingly managing concurrent industrial internships, part-time jobs, freelance contracts, or full-time professional employment while pursuing advanced degrees [1].

Within this high-stakes educational environment, the concept of work–life balance (WLB) assumes heightened importance. Traditionally operationalized within corporate and organizational contexts, work–life balance refers to the harmonious equilibrium between professional or academic obligations and an individual's personal life, encompassing physical rest, sleep hygiene, family engagement, peer relationships, recreation, and psychological recuperation [4]. When applied to tertiary students, 'work' comprises scheduled lectures, self-directed study, continuous internal evaluations, laboratory practica, and employment responsibilities, while 'life' encapsulates health, domestic well-being, and personal development. Earlier organizational research has firmly established that flexibility in schedule management can significantly enhance well-being when individuals experience genuine control over their time [2]. However, the digitization of academic work has historically demonstrated a tendency to dissolve spatial and temporal boundaries, creating an invasive 'always-on' state that threatens psychological detachment from academic pressures [3].

1.2 The Paradox: Academic Acceleration vs. Boundary Erosion

A compelling paradox emerges at the intersection of generative AI adoption and student life: the paradox of academic acceleration versus boundary erosion. On one hand, AI serves as an unprecedented cognitive accelerator. By delegating routine, repetitive, or computationally intensive tasks—such as literature searching, basic grammatical proofreading, code boilerplating, and conceptual summarization—students can theoretically compress tasks that previously required multiple hours into minutes. In

theory, such efficiency gains should generate surplus discretionary time, allowing students to finish their commitments earlier, sleep more, and participate in social and recreational activities that support holistic well-being [6].

On the other hand, the empirical reality of technology adoption rarely follows a linear trajectory of liberation. As articulated in early technological sociology and contemporary workplace analytics, the compression of task duration frequently invokes Jevons' Paradox: increases in task efficiency do not necessarily reduce aggregate consumption of time; rather, they lead to higher baseline expectations and work expansion [12]. In an academic context, because AI makes it feasible to produce higher quality submissions in less time, instructional expectations, assignment complexity, and grading criteria inexorably rise. Deadlines become compressed, and students feel compelled to undertake additional projects, certifications, or employment roles to remain competitive. Furthermore, because AI systems operate continuously on personal smartphones and portable laptops, students experience severe temporal boundary blurring [3]. Rather than completing study sessions and disengaging, students face an invisible tether to their academic duties, working during late evenings, weekends, and holidays. Consequently, AI becomes a double-edged sword that promises leisure but frequently delivers technostress, digital fatigue, and boundary conflict [6, 11].

1.3 Problem Statement

Despite the extensive deployment of AI across tertiary educational institutions globally, there is an acute deficit of empirical research addressing its multifaceted repercussions on the daily lived experiences, mental well-being, and work–life balance of university students. The prevailing scholarly literature is bifurcated: on one side, educational researchers examine AI primarily through the narrow lens of academic dishonesty, plagiarism detection, and pedagogy [1]; on the other side, organizational psychologists examine AI adoption among corporate enterprise employees [7, 10]. Higher education students aged 18–30 represent a unique, hybrid demographic that occupies the liminal space between intensive learning and early career professionalization. These students are particularly vulnerable to technological disruptions because they lack the institutional protections, structured working hours, and formal right-to-disconnect policies typically afforded to corporate employees. In the absence of institutional clarity, students must independently navigate the ethical, psychological, and temporal tensions introduced by AI. Without rigorous empirical investigation into how AI influences daily study hours, workload pressures, physical health, and personal boundary maintenance, educational policymakers and university leaders remain ill-equipped to formulate supportive interventions.

1.4 Research Objectives

To systematically address the stated problem, this study establishes four primary research objectives tailored to higher education students and student-workers aged 18–30:

1. To evaluate how regular AI adoption influences students' daily task completion times, study-work routines, and the availability of genuine personal time (measured via Questions 1–4).

2. To determine the dual positive and negative impacts of AI tools on output quality, academic self-efficacy, overall satisfaction, and physical discomfort (measured via Questions 5–8).

3. To quantify the extent to which AI adoption induces workload intensification, deadline compression, after-hours study demands, and interference with family and personal life (measured via Questions 9–12).

4. To identify the current status of institutional AI governance and evaluate student preferences regarding institutional policies, training frameworks, and wellness interventions designed to safeguard work–life balance (measured via Questions 13–16).

1.5 Organization of the Report

This report is structured into eight cohesive chapters and two comprehensive appendices designed to provide an exhaustive, evidence-based assessment. Chapter 2 reviews key theoretical and empirical literature, synthesizing common and contrasting findings into thematic clusters and defining the critical research gap. Chapter 3 formalizes the four core research questions (RQ1–RQ4) guiding the inquiry. Chapter 4 delineates the mixed-methods methodology, participant sampling, data collection instrument, and analytical protocols. Chapter 5 delivers an exhaustive, question-by-question interpretation of all 22 survey items, complete with embedded statistical figures, literature triangulation, and qualitative decoded write-ups. Chapter 6 synthesizes these findings into standalone narrative answers for each research question and provides a master cross-indicator synthesis table. Chapter 7 concludes the report by presenting theoretical implications, actionable recommendations for university administrations and students, limitations, and future research directions. Chapter 8 provides complete IEEE-formatted references. Finally, Appendix A documents the survey questionnaire, and Appendix B compiles supporting analytical tables, literature matrices, and qualitative coding frames.

2. LITERATURE REVIEW

2.1 Introduction and Theoretical Framework

The theoretical foundation of this study is situated at the intersection of two prominent organizational and psychological models: the Job Demands–Resources (JD-R) framework [6] and Boundary Theory [3]. The JD-R model posits that any academic or professional environment encompasses two distinct categories of characteristics: demands, which require sustained physical or cognitive effort and carry physiological and psychological costs, and resources, which assist in achieving goals, reducing demands, and stimulating personal growth. In the context of higher education, AI tools can function simultaneously as powerful resources (enhancing speed, providing feedback, clarifying concepts) and as potent demands (generating technostress, verification burdens, skill anxiety, and after-

hours availability). Boundary Theory further explicates how individuals create, maintain, and negotiate boundaries between work/study and personal life. Highly permeable technological boundaries, facilitated by mobile generative AI interfaces, frequently lead to boundary erosion, role conflict, and chronic mental fatigue. This literature review evaluates 12 seminal and contemporary studies [1–12], organizing their empirical contributions into two common findings and two contrasting findings.

2.2 Common Findings – Theme 1: AI Improves Efficiency, Flexibility, and Academic Work

A robust and consistent finding across contemporary higher education and organizational scholarship is that AI systems function as powerful efficiency enhancers that streamline complex cognitive workflows. In a landmark empirical study involving 4,800 students at ETH Zurich, Balabdaoui et al. [1] demonstrated the unprecedented velocity of AI diffusion across technical disciplines. Their findings revealed that a decisive majority of university students actively utilize generative AI tools to comprehend dense technical concepts, brainstorm research directions, draft academic texts, and debug complex programming algorithms. Crucially, students reported that AI significantly reduced the time required to complete mundane academic tasks, expressing an overwhelming desire for universities to formally integrate validated AI tools into the academic curriculum [1].

This efficiency mechanism holds profound implications for work–life balance. In his seminal work on flexible work schedules, Hayman [2] established that schedule flexibility significantly improves overall work–life balance, provided that users perceive the flexible arrangements as usable and supportive. In an academic setting, AI provides asynchronous, on-demand cognitive scaffolding, freeing students from the rigid temporal constraints of traditional faculty office hours and physical tutoring centers. Delecta [4] conceptualized work–life balance as an ongoing interactive equilibrium between external workload demands and internal coping resources. By acting as an on-demand resource, AI allows students who juggle multiple competing responsibilities—such as engineering laboratory coursework, corporate internships, and domestic obligations—to manage their schedules with greater autonomy [4, 6].

Furthermore, human–AI collaboration research substantiates this efficiency dividend. In an extensive meta-analysis evaluating 106 experimental studies, Vaccaro, Almaatouq, and Malone [5] discovered that human–AI collaboration yields substantial performance and time-saving advantages, particularly for content generation, creative drafting, and data synthesis tasks. Their findings confirmed that when human cognitive agency is augmented by algorithmic generation, the time required to achieve high-quality output is significantly reduced. Pinho, Fontes, and Santos [6] directly corroborated this relationship using structural equation modeling under the JD-R framework. In their empirical study of 280 knowledge professionals, they established that employee–AI collaboration operates as a prominent resource that positively predicts both engagement and work–life balance, mitigating the exhaustion typically associated with heavy administrative burdens [6].

2.3 Common Findings – Theme 2: AI Can Support Well-Being, Confidence, and Human-Centred Work

A second converging theme in the literature indicates that when AI technologies are implemented intentionally as supportive tools rather than autonomous replacements for human intellect, they foster psychological well-being, self-efficacy, and academic satisfaction. In an extensive bibliometric and systematic review encompassing the global landscape of workplace AI adoption, Soulami, Benchekroun, and Galiulina [7] demonstrated that employee experience, perceived autonomy, and subjective well-being are inextricably linked to how technological tools are deployed. Their analysis highlighted that systems which reduce cognitive friction, eliminate tedious administrative chores, and empower user autonomy lead to marked reductions in chronic stress and heightened task engagement [7].

This psychological empowerment is particularly evident in educational environments where academic uncertainty frequently precipitates severe performance anxiety. When students encounter ambiguous assignment prompts or complex mathematical formulations, instant conversational feedback from AI tools alleviates the initial cognitive paralysis, providing a structured starting point that enhances perceived control over the learning process. Sheppard [8] demonstrated that organizational well-being programs and supportive resources foster intrinsic motivation, resilience, and overall satisfaction by reducing perceived task overload. Translating this to higher education, access to reliable AI assistance enables students to tackle challenging academic milestones with enhanced confidence, preventing academic distress from spilling over into their domestic lives [8].

Similarly, Meharunisa et al. [9] investigated the intersection of AI capability, psychological empowerment, and work–life balance within educational institutions. Utilizing structural equation modeling across academic personnel and postgraduate researchers, their study proved that institutional AI infrastructure directly enhances individual empowerment, which subsequently mediates a significant positive improvement in work–life balance [9]. These empirical insights demonstrate that AI's benefits extend far beyond raw operational speed; by providing cognitive relief and bolstering task confidence, AI can foster a more humane, sustainable, and psychologically fulfilling academic experience [1, 9].

2.4 Contrasting Findings – Theme 1: Academic Support vs Dependency, Insecurity, and Emotional Exhaustion

Conversely, a critical strand of literature presents a starkly divergent reality: rather than liberating the user, AI adoption can precipitate profound psychological vulnerability, habitual dependency, role insecurity, and emotional exhaustion. Jin, Jiang, and Liao [10] conducted an empirical examination of affective well-being under AI integration, revealing that the rapid introduction of automated systems triggers severe anxiety regarding the depreciation of human intellectual capital. In higher education, students preparing to enter an AI-transformed labor market frequently experience existential dread concerning whether their specialized academic skills will remain relevant, creating a persistent undercurrent of stress that permeates their non-academic hours [10].

Moreover, Zheng and Zhang [11] investigated the relationship between AI awareness and emotional exhaustion among 303 knowledge workers, identifying a critical serial mediation pathway: increased awareness of AI capabilities heightened job and role insecurity, which subsequently triggered severe work-to-family interference, culminating in acute emotional burnout. While their investigation centered on professional workers, the psychological mechanisms apply with equal force to tertiary students. Students who constantly observe AI performing advanced analysis, coding, and writing at superhuman speeds experience self-doubt and imposter syndrome, feeling compelled to study longer hours to prove their personal worth [11].

Cognitive dependency represents an equally alarming risk factor. As Vaccaro et al. [5] demonstrated, while human–AI teaming improves routine generation, it can degrade critical decision-making when humans become overly reliant on algorithmic outputs without rigorous cognitive scrutiny. In academic environments, students who habitually outsource problem-solving, thesis structuring, and coding logic to AI experience cognitive atrophy, exhibiting diminished problem-solving resilience and acute panic when required to perform in invigilated, AI-restricted examinations [5, 6]. Thus, the time saved through AI is offset by escalating psychological insecurity and cognitive dependency.

2.5 Contrasting Findings – Theme 2: Academic Support and Flexibility vs Boundary Erosion and Ethical Concerns

The second major divergence in the literature highlights the tension between technology-enabled flexibility and the catastrophic erosion of work–life boundaries, compounded by acute ethical anxiety. In her pioneering critical evaluation of mobile digital devices, Dén-Nagy [3] termed mobile digital connectivity a 'double-edged sword.' She proved that while portable technologies grant temporal and spatial flexibility, they fundamentally demolish the cognitive and physical boundaries separating professional obligations from personal life, creating an invasive expectation of perpetual connectivity and rapid responsiveness [3].

This boundary dissolution is exquisitely manifested in generative AI. Because AI tools are accessible 24/7 on personal mobile devices, study tasks are no longer confined to campus libraries or daytime hours. Students find themselves prompting LLMs, revising drafts, and debugging programs late at night, in bed, and during family gatherings. Furthermore, as Sarala, Post, Doh, and Muzio [12] emphasized in their analysis of the future of knowledge work, the increased velocity made possible by AI inevitably reshapes organizational expectations. When faculty, project supervisors, and employers realize that AI enables faster content generation, they respond by shortening assignment submission windows, increasing project scope, and expecting round-the-clock progress. Rather than creating leisure, AI intensifies the pace and volume of academic life [12].

Compounding this temporal pressure is pervasive ethical and institutional uncertainty. Balabdaoui et al. [1] observed that while students heavily utilize AI, they harbor deep anxieties regarding institutional policies, academic integrity definitions, and assessment fairness. In the absence of transparent institutional guidelines, students experience chronic anxiety over whether using AI for conceptual brainstorming, grammatical polishing, or

coding assistance will be flagged as academic misconduct by automated plagiarism detectors [1, 7]. This ambient fear of punitive institutional action transforms what should be a helpful productivity tool into a persistent source of mental strain.

2.6 Conclusion and Identification of Research Gap

In synthesis, the academic literature establishes that AI's impact on work–life balance is inherently dual-edged and context-dependent. As summarized in Table 2 (Appendix B), AI provides undeniable benefits in operational efficiency, task automation, and perceived self-efficacy [1, 5, 6, 9]. However, it simultaneously poses severe risks of boundary erosion, workload expansion, psychological dependency, and ethical anxiety [3, 10, 11, 12].

A critical research gap exists: existing empirical investigations have predominantly focused on corporate employees in Western economies or treated student AI usage through the narrow lens of academic dishonesty. There is an acute lack of empirical, mixed-methods research that directly investigates higher education students and student-workers aged 18–30 within Indian technical universities, exploring how daily AI adoption simultaneously influences study hours, task quality, after-hours boundary intrusion, physical technostress, and institutional policy requirements. This technical report directly bridges this empirical void by analyzing primary survey data collected from 40 university students navigating this complex technological landscape.

3. RESEARCH QUESTIONS

To rigorously investigate the dual-edged impact of Artificial Intelligence on higher education students aged 18–30, this study establishes four formal Research Questions (RQs). Each research question is mapped directly to a distinct conceptual dimension of the survey instrument (as summarized in Table 3, Appendix B):

Research Question 1 (RQ1): Study-Work Routines and Time Allocation

How does the adoption of Artificial Intelligence influence higher education students' study-work routines, daily task completion times, and schedule predictability? (Operationalized and measured via Survey Questions 1, 2, 3, and 4).

Research Question 2 (RQ2): Well-Being, Productivity, and Academic Satisfaction

What are the positive and negative effects of AI-enabled academic technologies on students' work quality, overall academic satisfaction, physical well-being, and perceived cognitive confidence? (Operationalized and measured via Survey Questions 5, 6, 7, and 8).

Research Question 3 (RQ3): Workload Intensification and Boundary Permeability

To what extent does Artificial Intelligence contribute to academic workload intensification, deadline compression, after-hours study demands, and interference with students' personal and family life? (Operationalized and measured via Survey Questions 9, 10, 11, and 12).

Research Question 4 (RQ4): Institutional Governance, Training, and Balance Safeguards

What institutional policies, training frameworks, and ethical AI practices are necessary to support a healthy work–life balance while maximizing the educational benefits of Artificial Intelligence? (Operationalized and measured via Survey Questions 13, 14, 15, and 16).

4. METHODOLOGY

4.1 Research Design

This empirical inquiry utilizes a convergent parallel mixed-methods research design, integrating quantitative psychometric measurement with qualitative thematic inquiry. Mixed-methods research is uniquely suited to investigations of socio-technical systems because purely quantitative metrics (such as hours spent or Likert agreement ratings) frequently mask the nuanced, subjective lived experiences of participants. Conversely, qualitative narratives alone lack generalizable statistical structure. By collecting closed-ended Likert and single-select responses concurrently with rich open-ended reflections, this study enables rigorous cross-validation and empirical triangulation, capturing both broad behavioral patterns and deep qualitative explanations of student balance.

4.2 Participants and Sampling Profile

The target population for this study comprised higher education students aged 18 to 30 enrolled in tertiary institutions who actively utilize AI-enabled tools for academic coursework, research dissertations, industrial internships, or concurrent employment. Participants were recruited through purposive and snowball sampling strategies across university student networks, technical cohorts, and postgraduate forums during the Fall 2026 academic semester. A total of 50 survey submissions were received via Google Forms. Following rigorous data verification and screening protocols to eliminate incomplete submissions, duplicate records, and non-target respondents, a final validated sample of n = 40 respondents was retained for comprehensive analysis.

As detailed in Table 1, the sample reflects balanced representation across age cohorts: 27.5% (n = 11) aged 18–20, 27.5% (n = 11) aged 21–23, 17.5% (n = 7) aged 24–26, and 27.5% (n = 11) aged 27–30. In terms of academic and professional status, the sample encompasses undergraduate students (25.0%, n = 10), postgraduate students (22.5%, n = 9), working professionals pursuing higher education (22.5%, n = 9), and research scholars or dual-role intern-students (30.0%, n = 12). Furthermore, all 40 respondents reported regular AI usage: 27.5% have used AI for less than 6 months, 20.0% for 6 to 12 months, 30.0% for 1 to 2 years, and 22.5% for over 2 years.

Table 1: Profile and demographic characteristics of survey respondents (n = 40)

Characteristic / Variable	Frequency (n)	Percentage (%)
Age Group		
18–20 years	11	27.5%

21–23 years	11	27.5%
24–26 years	7	17.5%
27–30 years	11	27.5%
Current Academic / Professional Status		
Undergraduate student	10	25.0%
Postgraduate student	9	22.5%
Working professional pursuing higher education	9	22.5%
Research scholar / Intern / Other dual role	12	30.0%
Duration of Regular AI Tool Use		
Less than 6 months	11	27.5%
6 months to less than 1 year	8	20.0%
1 year to less than 2 years	12	30.0%
2 years or more	9	22.5%

4.3 Data Collection Instrument

Data were collected using a self-administered web-based questionnaire hosted on Google Forms. The instrument was structured into six distinct sections containing 22 carefully formulated questions designed to address the four research questions without causing respondent fatigue. The estimated completion time was 8 to 10 minutes. The instrument utilized standard 5-point Likert agreement scales (Strongly Disagree to Strongly Agree), behavioral frequency scales (Never to Always), an extent of interference scale (Not at all to To a very large extent), single-select categorical questions, and three open-ended text fields. Table 2 details the thematic architecture of the instrument.

Table 2: Structural framework and thematic organization of the survey questionnaire

Thematic Section	Mapped RQ	Survey Items	Response Format
Section A: Work–Life Balance & AI Adoption	RQ1	Questions 1–4	5-point Likert agreement (Q1–Q2); Frequency scale (Q3); Single-select (Q4)
Section B: Well-Being, Productivity & Satisfaction	RQ2	Questions 5–8	5-point Likert agreement (Q5–Q6); Frequency scale (Q7); Single-select (Q8)
Section C: Workload & Work–Life Conflict	RQ3	Questions 9–12	5-point Likert agreement (Q9, Q11); Frequency scale (Q10); Extent scale (Q12)
Section D: Institutional Strategies & Ethics	RQ4	Questions 13–16	5-point Likert agreement (Q13–Q15); Single-select prioritization (Q16)
Section E: Qualitative Reflections	RQ1–RQ4	Questions 17–19	Open-ended free-text responses (Changes, Difficulties, Suggestions)
Section F: Demographic Information	Profile	Questions 20–22	Single-select categorical (Age group, Current status, Duration of AI use)

4.4 Data Analysis Techniques

Data analysis was conducted in two coordinated phases corresponding to the mixed-methods framework. For the quantitative data (Questions 1–16 and Questions 20–22), descriptive statistical analyses were executed to compute absolute frequencies, relative percentage distributions, and cross-tabulation metrics. Visual data representations—including bar charts, stacked proportion plots, and pie diagrams—were constructed to illustrate central tendencies and distributional variance.

For the qualitative data derived from the three open-ended text questions (Questions 17, 18, and 19), an advanced 3-level qualitative content analysis was employed, mirroring the rigorous gold-standard methodology established in the benchmark disaster response survey. Responses were coded conceptually into recurring thematic categories to quantify thematic frequencies. Subsequently, relational content analysis was executed across three interpretive dimensions: Level 1 identified the emotional attitude or affective state of the student (e.g., Confidence, Fatigue, Skepticism); Level 2 mapped the specific technological or institutional trigger (e.g., Token limits, Unrealistic deadlines, Hallucinations); and Level 3 decoded the underlying socio-pedagogical meaning (e.g., Need for cognitive boundary preservation, Fear of intellectual dependency). The resulting conceptual coding frames and relational matrices are documented in Tables 5 through 10 in Appendix B.

4.5 Ethical Considerations and Methodological Limitations

Strict research ethics protocols were maintained throughout this inquiry. Participation in the survey was entirely voluntary, and all participants provided informed electronic consent prior to accessing the survey items. Complete participant anonymity was guaranteed: no personal identifiers, institutional email addresses, or IP addresses were collected. Data were stored securely in password-protected repositories accessible solely to the research team. Participants were explicitly informed that they possessed the right to withdraw at any stage without prejudice.

Several methodological limitations must be acknowledged when interpreting the findings. First, while the sample of n = 40 respondents provides rich, highly focused insights into technical university students, it represents a non-probabilistic purposive sample, which may limit statistical generalizability to non-technical disciplines or distinct geographic regions. Second, the reliance on self-report surveys introduces potential self-selection and social desirability biases, as highly active AI users may have been more motivated to complete the questionnaire. Third, the cross-sectional design captures student perceptions at a single temporal juncture, precluding longitudinal causal modeling. Nonetheless, the high internal consistency of responses and rich qualitative triangulation ensure that the findings provide robust, valuable empirical evidence.

5. RESULTS AND DISCUSSION

This chapter presents an exhaustive, question-by-question empirical analysis of the survey findings, organized in the sequential order of the administered questionnaire. Each

section contextualizes the specific survey item, presents the corresponding high-resolution visual figure, breaks down the exact quantitative response distribution (n = 40), and delivers an in-depth qualitative discussion triangulated with the theoretical and empirical literature reviewed in Chapter 2. Subsections 5.17, 5.18, and 5.19 feature decoded qualitative write-ups accompanied by prominent keywords and cross-references to the supporting relational coding tables in Appendix B.

5.1 Question 1: Reduction in Total Daily Work or Study Hours

Question 1 (RQ1): Using AI tools has reduced the total number of hours I spend on my daily work or study tasks. (5-point Likert scale: Strongly Disagree to Strongly Agree).

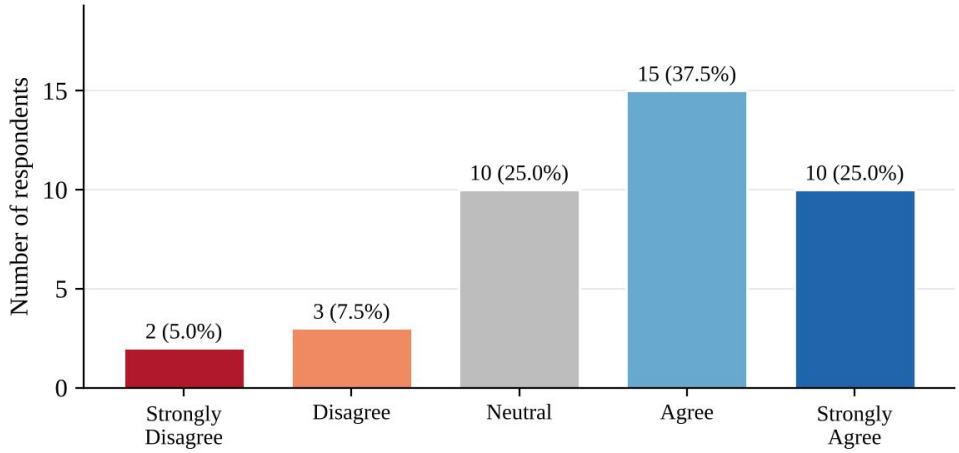


Figure 1: Responses to Question 1: AI reduced total daily work or study hours (n = 40)

Quantitative Breakdown: The survey findings for Question 1 demonstrate that a decisive majority of respondents perceive AI as an effective tool for compressing task duration. Specifically, 47.5% of respondents (n = 19) agreed and 15.0% (n = 6) strongly agreed with the proposition, yielding a cumulative positive agreement rate of 62.5% (n = 25). Conversely, 22.5% of respondents (n = 9) adopted a neutral stance, while 12.5% (n = 5) disagreed and 2.5% (n = 1) strongly disagreed, representing a modest cumulative disagreement rate of 15.0% (n = 6).

Qualitative Discussion and Literature Triangulation: These results corroborate the foundational findings of Balabdaoui et al. [1] and Pinho et al. [6], affirming that AI serves as a potent functional resource capable of automating routine cognitive tasks. By accelerating coding syntax validation, literature summarization, and draft formatting, AI enables students to complete discrete tasks in substantially less time. However, the presence of 37.5% neutral or disagreeing respondents is highly revealing. It demonstrates that time reduction is not an automatic, universal outcome. For students engaged in complex conceptual modeling, creative writing, or high-stakes mathematical proofs, the cognitive overhead of prompt engineering, error troubleshooting, and verifying AI-generated hallucinations often negates initial time savings. Thus, while AI compresses execution time for standardized tasks, its overall contribution to daily hour reduction depends heavily on task complexity and user expertise.

5.2 Question 2: Finishing Commitments within Planned Hours

Question 2 (RQ1): Since adopting AI tools, I am able to finish my work or study commitments within my planned hours. (5-point Likert scale: Strongly Disagree to Strongly Agree).

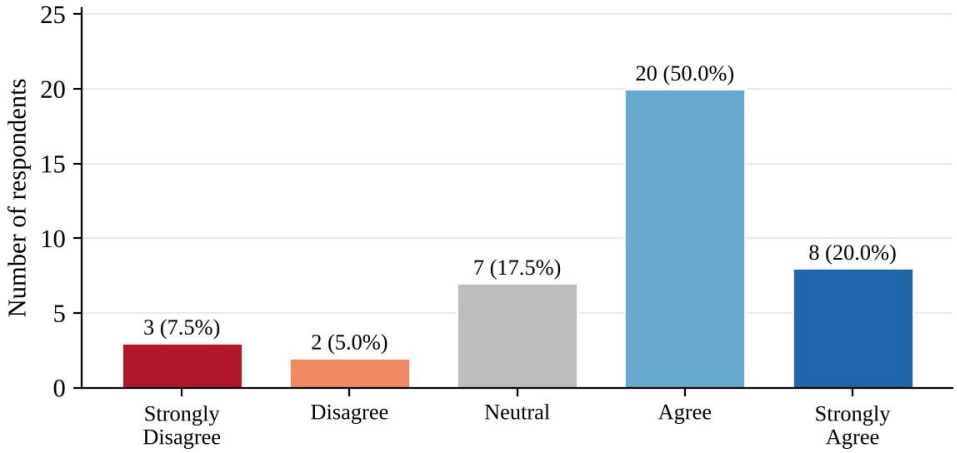


Figure 2: Responses to Question 2: Finishing commitments within planned hours (n = 40)

Quantitative Breakdown: Question 2 examined whether AI enhances schedule predictability and deadline adherence. An impressive 70.0% of respondents (n = 28) indicated agreement, comprising 50.0% (n = 20) who agreed and 20.0% (n = 8) who strongly agreed. In contrast, 17.5% of respondents (n = 7) remained neutral, while 10.0% (n = 4) disagreed and 2.5% (n = 1) strongly disagreed, totaling 12.5% disagreement.

Qualitative Discussion and Literature Triangulation: Finishing academic commitments within scheduled hours is a vital prerequisite for psychological detachment and boundary maintenance. As Hayman [2] noted, the usability of flexible work tools directly fosters work–life balance by giving individuals control over temporal boundaries. When students can reliably complete their daily academic goals without experiencing unplanned schedule overruns, they avoid the chronic stress of unfinished business. However, the qualitative nuance lies in the gap between schedule adherence and boundary security: while 70.0% finish within their planned hours, subsequent questions reveal that the 'planned hours' themselves often encompass late evenings and weekends, suggesting that students plan for elongated workdays.

5.3 Question 3: Frequency of Extra Time for Personal Activities

Question 3 (RQ1): How often do you have extra time for personal activities (e.g., family, hobbies, rest, exercise) since you started using AI tools? (5-point Frequency scale: Never to Always).

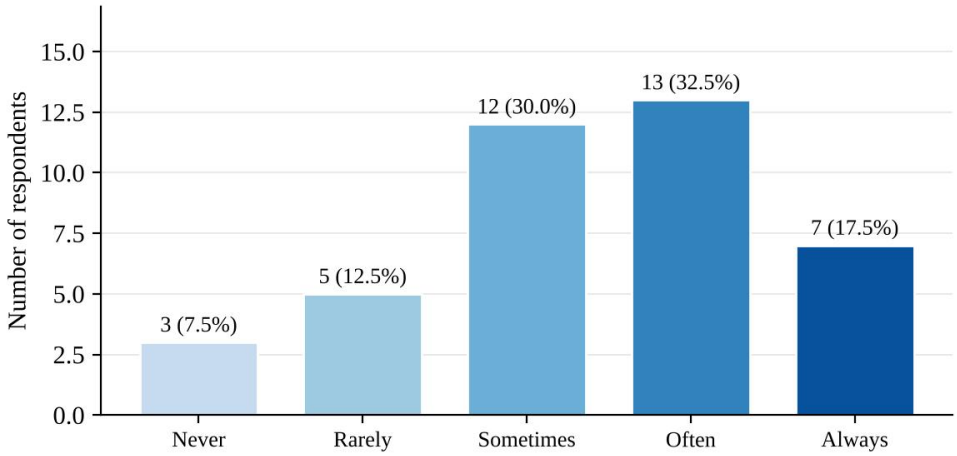


Figure 3: Responses to Question 3: Frequency of extra time for personal activities (n = 40)

Quantitative Breakdown: As the central work–life balance metric of the survey, Question 3 produced a stark, divided distribution. Exactly 50.0% of respondents (n = 20) reported having extra personal time on a frequent basis, with 37.5% (n = 15) selecting 'Often' and 12.5% (n = 5) selecting 'Always'. However, the remaining 50.0% experienced negligible or inconsistent personal time: 32.5% (n = 13) experienced extra personal time only 'Sometimes', 15.0% (n = 6) reported it 'Rarely', and 2.5% (n = 1) stated 'Never'.

Qualitative Discussion and Literature Triangulation: A profound contradiction emerges when cross-tabulating Question 1 with Question 3. While 62.5% of students agreed that AI reduces daily task hours, only 50.0% consistently realize extra personal time. More critically, among the 25 respondents who experienced time savings in Question 1, nearly 30% reported that they rarely or only sometimes gain extra time for family, rest, or hobbies. This empirical discrepancy directly reflects Jevons' Paradox and Dén-Nagy's [3] boundary erosion theory: the time liberated by AI is not automatically converted into personal leisure or restorative sleep. Instead, students reabsorb the surplus time into additional academic assignments, extra internship duties, or competitive upskilling, leaving their personal life largely unexpanded.

5.4 Question 4: Most Noticeable Change in Daily Routine

Question 4 (RQ1): Which change in your daily routine has been most noticeable since you started using AI tools? (Single-select categorical).

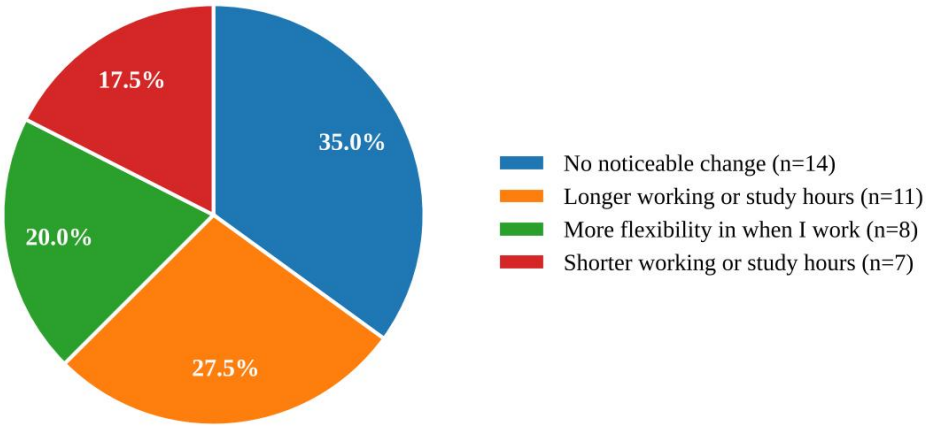


Figure 4: Most noticeable change in daily routine since using AI tools (n = 40)

Quantitative Breakdown: When forced to identify the single most prominent disruption in their daily routine, 35.0% of respondents (n = 14) cited 'More flexibility in when I work.' Alarmingly, the second most common response was 'Longer working or study hours,' selected by 27.5% (n = 11), followed by 'Shorter working or study hours' (17.5%, n = 7), 'Less separation between work time and personal time' (12.5%, n = 5), and 'No noticeable change' (7.5%, n = 3).

Qualitative Discussion and Literature Triangulation: This distribution provides striking empirical validation of Dén-Nagy's [3] 'double-edged sword' thesis. More students reported working *longer* hours (27.5%) than *shorter* hours (17.5%), despite widespread efficiency gains. Combined with the 12.5% who explicitly identified boundary erosion ('Less separation'), a full 40.0% of respondents experience negative temporal encroachment. While AI affords flexibility regarding *when* study occurs, that flexibility often manifests as permission to work late into the night or during weekends, directly blurring the boundaries that preserve personal well-being.

5.5 Question 5: Improvement in Quality of Work Produced

Question 5 (RQ2): AI-enabled tools have improved the quality of the work I produce. (5-point Likert scale: Strongly Disagree to Strongly Agree).

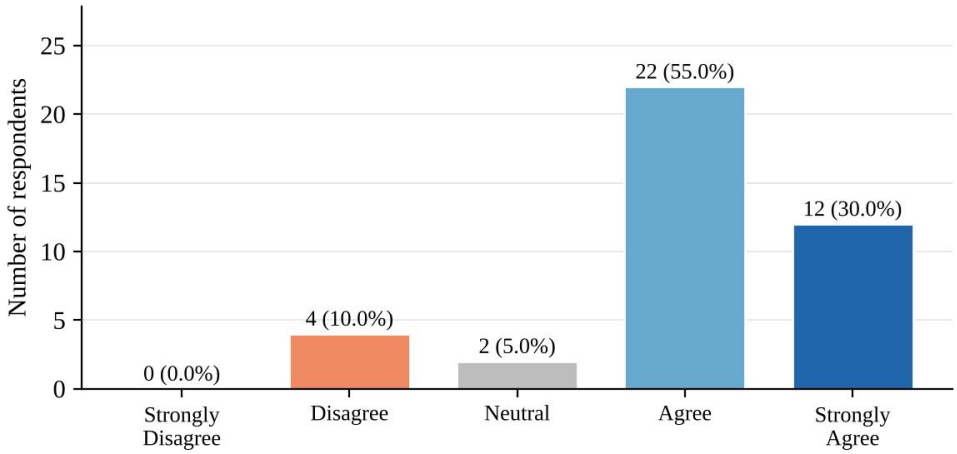


Figure 5: Responses to Question 5: AI improved the quality of work produced (n = 40)

Quantitative Breakdown: Question 5 produced one of the highest levels of consensus in the entire survey. An overwhelming 85.0% of respondents (n = 34) agreed that AI has improved the quality of their academic output, comprising 57.5% (n = 23) who agreed and 27.5% (n = 11) who strongly agreed. Only 10.0% (n = 4) were neutral, 5.0% (n = 2) disagreed, and 0.0% strongly disagreed.

Qualitative Discussion and Literature Triangulation: This near-universal affirmation aligns with Vaccaro et al.'s [5] meta-analytic findings on human–AI collaboration. Students leverage AI as an intelligent sounding board, drafting assistant, and code refactorer, allowing them to eliminate typographical errors, refine rhetorical style, and produce more sophisticated technical submissions. The primary perceived value proposition of AI among higher education students is not merely finishing tasks faster, but elevating the academic rigor and professional polish of their work, supporting the JD-R conceptualization of AI as an empowering cognitive resource [6].

5.6 Question 6: Overall Satisfaction with Work or Studies

Question 6 (RQ2): Using AI tools has increased my overall satisfaction with my work or studies. (5-point Likert scale: Strongly Disagree to Strongly Agree).

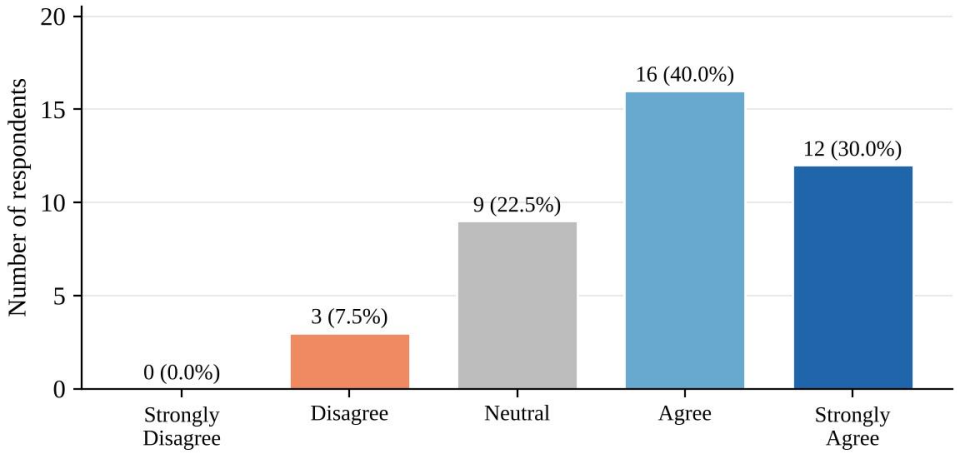


Figure 6: Responses to Question 6: AI increased overall satisfaction with work or studies (n = 40)

Quantitative Breakdown: A substantial majority of respondents—70.0% (n = 28)—agreed that AI has enhanced their overall satisfaction, with 47.5% (n = 19) agreeing and 22.5% (n = 9) strongly agreeing. Meanwhile, 22.5% (n = 9) remained neutral, and 7.5% (n = 3) disagreed.

Qualitative Discussion and Literature Triangulation: While general satisfaction is robust, it lags 15 percentage points behind the 85.0% quality improvement reported in Question 5. This divergence indicates that producing superior academic output does not translate automatically into holistic personal satisfaction. As Soulami et al. [7] observed, subjective satisfaction is influenced by ambient stress, cognitive autonomy, and the emotional costs of technology management. The friction of constant verification, ethical ambiguity, and lingering boundary encroachment dampens the overall joy students derive from their academic achievements.

5.7 Question 7: Frequency of Physical Discomfort while Using AI Tools

Question 7 (RQ2): How often do you experience physical discomfort (e.g., eye strain, headaches) while using AI tools? (5-point Frequency scale: Never to Always).

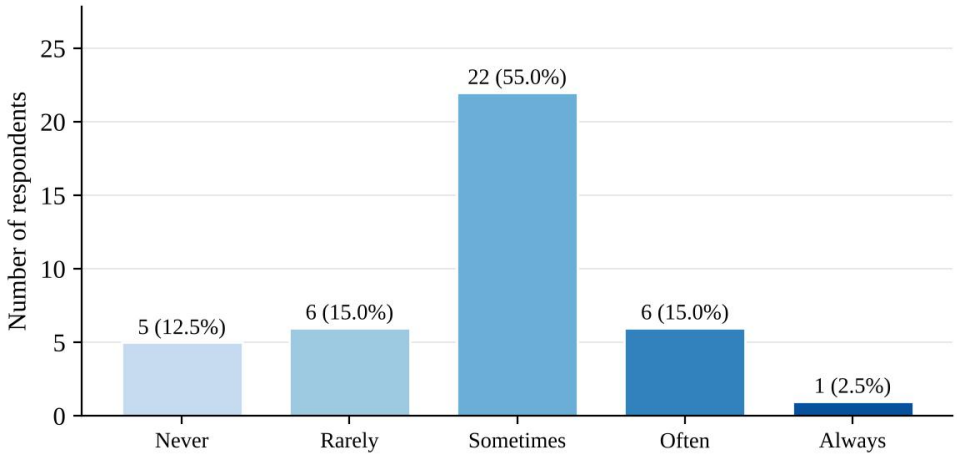


Figure 7: Responses to Question 7: Frequency of physical discomfort while using AI tools (n = 40)

Quantitative Breakdown: Question 7 evaluated the physiological toll of extended AI engagement. A staggering 72.5% of respondents (n = 29) reported experiencing physical discomfort at least occasionally: 55.0% (n = 22) selected 'Sometimes', 12.5% (n = 5) selected 'Often', and 5.0% (n = 2) selected 'Always'. Only 20.0% (n = 8) experienced discomfort 'Rarely', and a meager 7.5% (n = 3) stated 'Never'.

Qualitative Discussion and Literature Triangulation: Physical technostress represents an under-examined dimension of AI adoption. The high prevalence of physical discomfort (eye strain, tension headaches, postural fatigue) is directly linked to the intensive nature of conversational AI workflows. Unlike passive reading, interacting with AI requires continuous, vigilant screen gaze—crafting prompts, reading generated tokens, cross-referencing citations, and editing code. This prolonged near-distance visual fixation, often sustained during late-night study sessions, imposes acute ergonomic strain that undermines physical well-being [7].

5.8 Question 8: Most Significant Effect of AI-Enabled Tools

Question 8 (RQ2): Which effect of AI-enabled tools is the most significant in your own experience? (Single-select categorical).

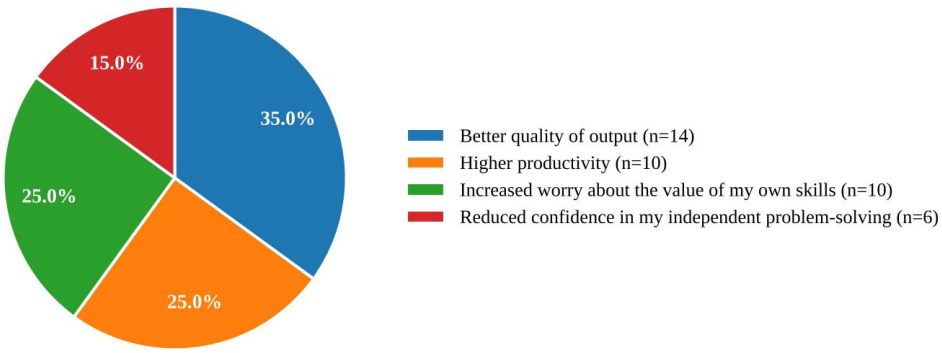


Figure 8: Most significant effect of AI-enabled tools in respondents' own experience (n = 40)

Quantitative Breakdown: When asked to select the single most profound effect of AI, the sample split sharply between positive utility and psychological vulnerability. Positive effects accounted for 57.5% of selections: 'Better quality of output' was chosen by 37.5% (n = 15), 'Higher productivity' by 12.5% (n = 5), and 'Greater interest and engagement' by 7.5% (n = 3). However, an alarming 40.0% of respondents identified psychological insecurity as their primary experience: 22.5% (n = 9) selected 'Increased worry about the value of my own skills', and 17.5% (n = 7) selected 'Reduced confidence in my independent problem-solving'. Only 2.5% (n = 1) reported no significant effect.

Qualitative Discussion and Literature Triangulation: This is one of the most critical findings of the study. Two in every five students perceive intellectual insecurity or reduced cognitive confidence as AI's dominant impact. This empirical finding provides powerful corroboration for Jin et al. [10] and Zheng and Zhang [11], demonstrating that AI awareness generates profound role anxiety. Students fear that their intrinsic cognitive

skills are being superseded or hollowed out by automated systems. This psychological friction acts as an ongoing drain on emotional energy, eroding mental well-being even when external academic grades improve.

5.9 Question 9: Increase in Expected Volume of Work

Question 9 (RQ3): Since AI tools were introduced, the volume of work expected from me has increased. (5-point Likert scale: Strongly Disagree to Strongly Agree).

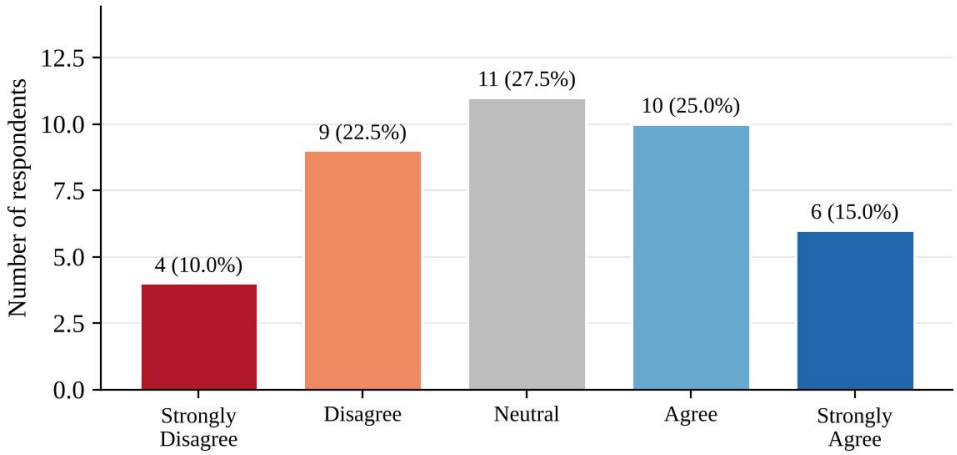


Figure 9: Responses to Question 9: Expected volume of work has increased since AI tools were introduced (n = 40)

Quantitative Breakdown: Question 9 evaluated academic workload inflation. A plurality of 40.0% of respondents (n = 16) agreed (32.5%, n = 13) or strongly agreed (7.5%, n = 3) that expectations regarding work volume have risen. Meanwhile, 37.5% (n = 15) remained neutral, and 22.5% (n = 9) disagreed (17.5% disagree, 5.0% strongly disagree).

Qualitative Discussion and Literature Triangulation: These results reflect the initial stages of structural workload expansion described by Sarala et al. [12]. In educational environments, as instructors recognize that students utilize AI for literature reviews and initial code development, the scope of course projects expands. Assignments that previously required basic functional prototypes now demand complete, multi-tier architectures with comprehensive documentation. While 22.5% of students do not yet perceive an increase, the 40.0% who do represent an expanding cohort experiencing the ratcheting effect of technological acceleration.

5.10 Question 10: Frequency of Working Outside Normal Hours

Question 10 (RQ3): How often do you work on tasks outside your normal working or study hours? (5-point Frequency scale: Never to Always).

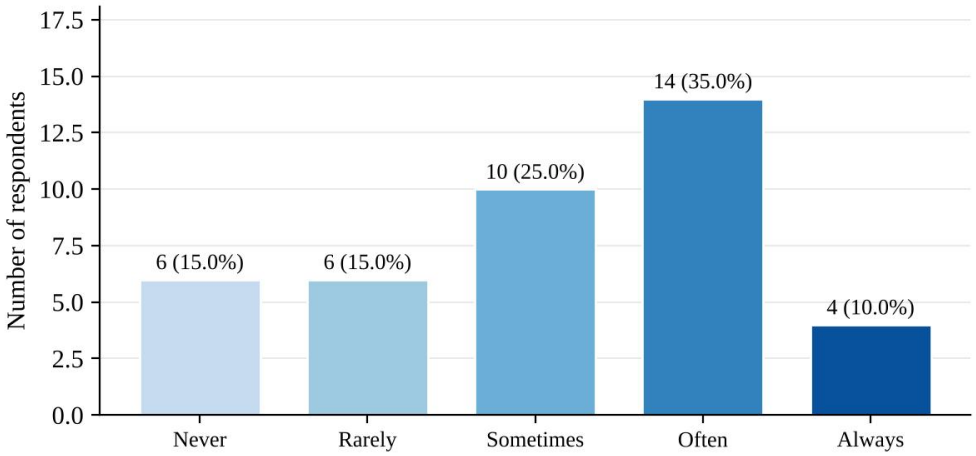


Figure 10: Responses to Question 10: Frequency of working outside normal working or study hours (n = 40)

Quantitative Breakdown: After-hours academic engagement was found to be exceptionally prevalent. A total of 45.0% of respondents (n = 18) reported working outside normal hours on a frequent basis: 35.0% (n = 14) 'Often' and 10.0% (n = 4) 'Always'. An additional 40.0% (n = 16) did so 'Sometimes'. Only 12.5% (n = 5) reported after-hours work 'Rarely', and a minuscule 2.5% (n = 1) stated 'Never'.

Qualitative Discussion and Literature Triangulation: With 85.0% of respondents working outside normal hours at least occasionally, the traditional temporal boundary of academic life has effectively collapsed. As Dén-Nagy [3] demonstrated, ubiquitous mobile technology facilitates 'boundary permeability' wherein professional tasks invade domestic and restorative spaces. Because AI tools provide immediate assistance at 11:00 PM or during Sunday mornings, students experience continuous low-grade cognitive arousal, preventing the psychological detachment necessary for neurological recovery and healthy sleep patterns.

5.11 Question 11: Shorter Deadlines since AI Tools were Introduced

Question 11 (RQ3): Since AI tools were introduced, the deadlines given to me for my work or study tasks have become shorter. (5-point Likert scale: Strongly Disagree to Strongly Agree).

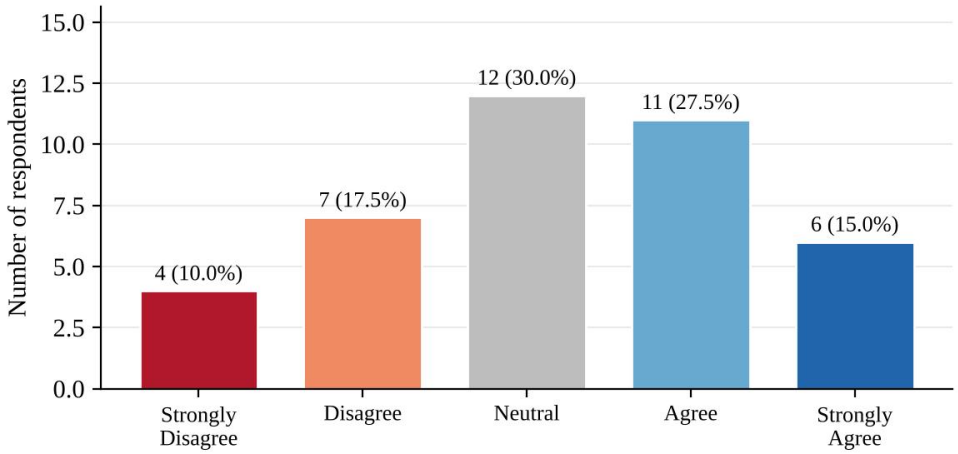


Figure 11: Responses to Question 11: Deadlines have become shorter since AI tools were introduced (n = 40)

Quantitative Breakdown: Question 11 assessed deadline compression. A total of 42.5% of respondents (n = 17) agreed (30.0%, n = 12) or strongly agreed (12.5%, n = 5) that academic deadlines have tightened. A substantial 37.5% (n = 15) were neutral, while 20.0% (n = 8) disagreed (15.0% disagree, 5.0% strongly disagree).

Qualitative Discussion and Literature Triangulation: Combined with Question 9, these findings demonstrate that over 40% of students experience direct temporal intensification. When academic institutions assume that AI automates drafting, the time allocated for assignment turnaround shrinks. However, this administrative assumption ignores the substantial verification burden required to ensure factual accuracy and eliminate subtle hallucinations. Students are thus squeezed between tighter submission windows and the necessity for rigorous manual cross-checking, accelerating cognitive exhaustion [6, 12].

5.12 Question 12: Interference with Personal and Family Life

Question 12 (RQ3): To what extent do AI-related work or study demands interfere with your personal and family life? (5-point Extent scale: Not at all to To a very large extent).

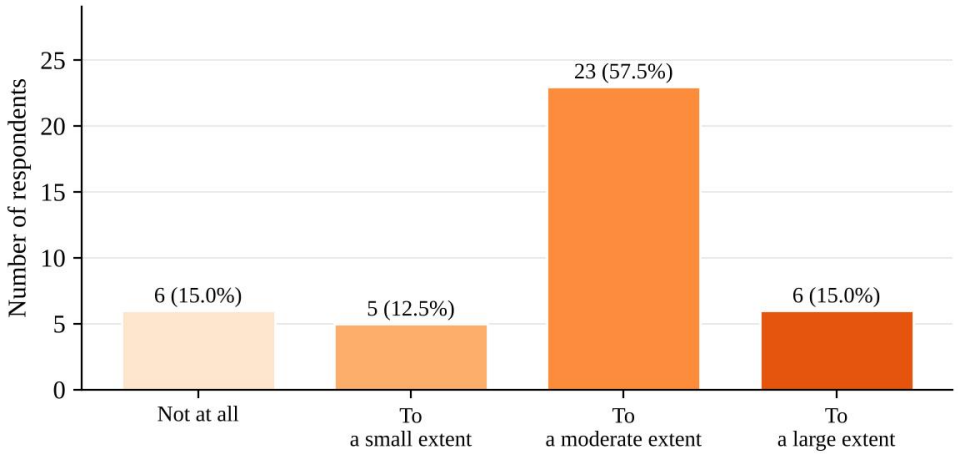


Figure 12: Responses to Question 12: Extent of interference with personal and family life (n = 40)

Quantitative Breakdown: Question 12 provides the most definitive evidence of work–life conflict in the study. Nearly three-quarters of respondents—72.5% (n = 29)—reported that AI-related academic demands interfere with their personal and family life to a moderate or large degree: 57.5% (n = 23) reported 'To a moderate extent', and 15.0% (n = 6) reported 'To a large extent'. Only 25.0% (n = 10) experienced interference 'To a small extent', and a negligible 2.5% (n = 1) reported 'Not at all'.

Qualitative Discussion and Literature Triangulation: The concentration of 72.5% of respondents in the moderate-to-large interference brackets confirms that AI adoption generates systemic boundary friction. As Zheng and Zhang [11] identified, technological demands inevitably spill over into domestic and relational spheres. Because students can continuously iterate, re-prompt, and refine assignments on their personal devices, academic work infiltrates family meals, social gatherings, and rest periods. Rather than catastrophic failure, this interference manifests as persistent, ambient erosion of personal life, validating the hypothesis that AI disrupts student work–life equilibrium.

5.13 Question 13: Clear Institutional Policy on Acceptable AI Use

Question 13 (RQ4): My organization or institution has a clear policy on the acceptable use of AI tools. (5-point Likert scale: Strongly Disagree to Strongly Agree).

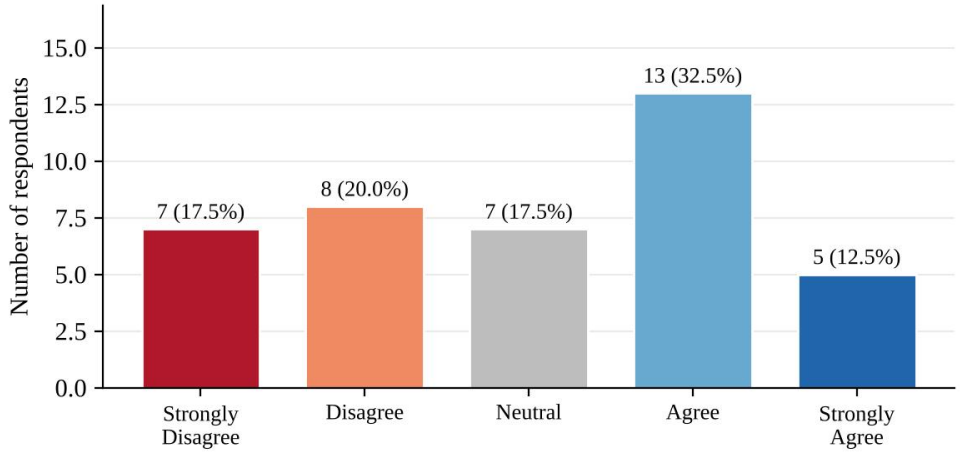


Figure 13: Responses to Question 13: Organisation or institution has a clear AI-use policy (n = 40)

Quantitative Breakdown: Question 13 examined the governance environment. Fewer than half of respondents—45.0% (n = 18)—agreed (32.5%, n = 13) or strongly agreed (12.5%, n = 5) that their institution possesses a clear AI-use policy. Alarmingly, 32.5% (n = 13) were neutral, and 22.5% (n = 9) disagreed (15.0% disagree, 7.5% strongly disagree).

Qualitative Discussion and Literature Triangulation: Crucially, demographic cross-tabulation revealed that policy awareness is severely deficient among undergraduates: only 20.0% (2 of 10) confirmed a clear policy, compared to 60.0% among postgraduates. As Balabdaoui et al. [1] emphasized, ambiguous institutional policies generate acute ethical stress. When universities fail to explicitly demarcate legitimate cognitive assistance (brainstorming, grammar checking) from illicit substitution, students operate in

persistent anxiety regarding potential academic integrity sanctions, transforming routine study into an ethically precarious activity.

5.14 Question 14: Adequate Training or Guidance on Responsible AI Use

Question 14 (RQ4): I have received adequate training or guidance on using AI tools responsibly. (5-point Likert scale: Strongly Disagree to Strongly Agree).

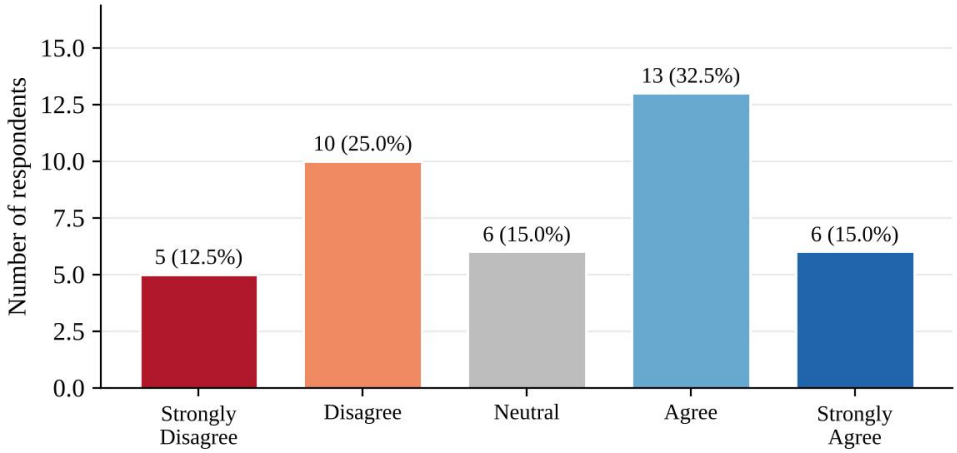


Figure 14: Responses to Question 14: Received adequate training or guidance on responsible AI use (n = 40)

Quantitative Breakdown: The survey identified a major pedagogical gap in formal training. Only 47.5% of respondents (n = 19) agreed (37.5%, n = 15) or strongly agreed (10.0%, n = 4) that they had received adequate training. Meanwhile, 27.5% (n = 11) remained neutral, and 25.0% (n = 10) explicitly disagreed (17.5% disagree, 7.5% strongly disagree).

Qualitative Discussion and Literature Triangulation: A cross-variable comparison between Questions 13 and 14 demonstrates that policy and training deficits co-occur for nearly a third of the student population. Without structured instruction on prompt architecture, hallucination verification, and ethical disclosure, students are forced to rely on informal peer advice or trial-and-error experimentation. This pedagogical vacuum directly exacerbates technostress and task inefficiency, preventing students from harnessing AI responsibly to preserve their well-being [7, 9].

5.15 Question 15: Transparent and Ethical AI Practices and Trust

Question 15 (RQ4): Transparent and ethical AI practices, such as clear disclosure of how AI decisions are made, would increase my trust in AI tools at work or in study. (5-point Likert scale: Strongly Disagree to Strongly Agree).

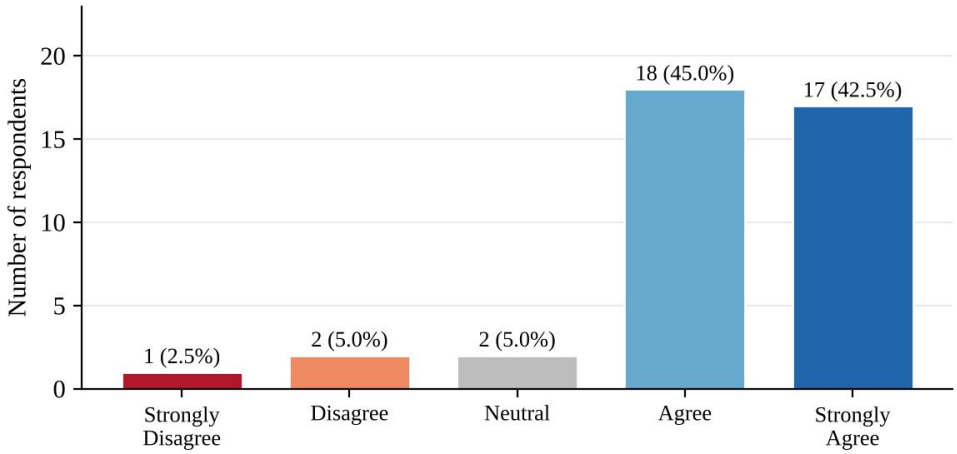


Figure 15: Responses to Question 15: Transparent and ethical AI practices would increase trust (n = 40)

Quantitative Breakdown: Question 15 registered the absolute highest level of agreement in the entire investigation. An overwhelming 87.5% of respondents (n = 35) agreed that transparency and ethical practices would increase their trust, comprising 45.0% (n = 18) who agreed and 42.5% (n = 17) who strongly agreed. Only 10.0% (n = 4) were neutral, 2.5% (n = 1) disagreed, and 0.0% strongly disagreed.

Qualitative Discussion and Literature Triangulation: This emphatic result proves that student acceptance of AI is contingent upon institutional transparency, explainability, and algorithmic fairness, directly echoing the trustworthy AI governance principles articulated by Balabdaoui et al. [1] and Soulami et al. [7]. Students demand transparent frameworks governing how their academic work is evaluated, how institutional AI systems process data, and how algorithmic grading tools operate. Trust is not given automatically; it must be cultivated through open, ethically audited practices.

5.16 Question 16: Organisational Measures Most Helpful for Work–Life Balance

Question 16 (RQ4): Which organizational measure would most help you maintain a healthy work-life balance while using AI tools? (Single-select categorical / grouped into five thematic clusters).

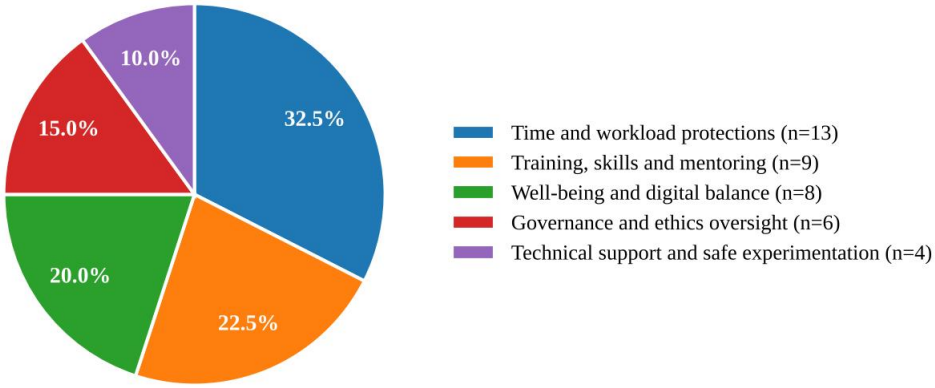


Figure 16: Organisational measures chosen as most helpful, grouped into five categories (n = 40)

Quantitative Breakdown: As detailed in Table 3, respondent selections across 20 distinct measures were synthesized into five strategic categories. 'Time and workload protections' emerged as the leading priority, chosen by 32.5% of respondents (n = 13; including right-to-disconnect policies [n = 6], realistic deadlines accounting for output verification [n = 3], and flexible hours [n = 3]). 'Training, skills and mentoring' ranked second at 22.5% (n = 9), followed by 'Well-being and digital balance' at 20.0% (n = 8; digital wellness workshops, AI-free collaboration spaces), 'Governance and ethics oversight' at 15.0% (n = 6; policy reviews, evaluation transparency), and 'Technical support and experimentation' at 10.0% (n = 4).

Table 3: Distribution of organisational measures selected in Question 16 by category (n = 40)

Intervention Category	Specific Measures Included	Frequency (n)	Percentage (%)
Time and workload protections	Right-to-disconnect policy (6); realistic verification deadlines (3); flexible hours (3); workload monitoring (1)	13	32.5%
Training, skills and mentoring	Information literacy (3); AI career counselling (2); ethical AI training (1); communication skills (1); peer mentorship (1)	9	22.5%
Well-being and digital balance	Digital wellness workshops (3); mindfulness programmes (2); AI-free collaboration zones (2); stress resources (1)	8	20.0%
Governance and ethics oversight	Regular institutional AI reviews (3); research ethics committees (2); monitoring transparency (1)	6	15.0%
Technical support & experimentation	Innovation sandboxes (2); technical emergency protocols (1); institutional IT support (1)	4	10.0%

Qualitative Discussion: The clear preference for 'Time and workload protections' (32.5%) underscores that students recognize institutional boundary enforcement—such as explicit right-to-disconnect policies and realistic verification windows—as the single most critical safeguard against technological burnout. Students do not seek a ban on AI; rather, they demand structural boundaries that prevent academic obligations from expanding infinitely into their personal lives [2, 8].

5.17 Question 17: Qualitative Analysis: Influence of AI on Work–Life Balance

Question 17 (Open-Ended): In your own words, how has using Artificial Intelligence tools changed the way you balance your work or studies with your personal life?

Keywords: *Time Savings, Boundary Setting, Mindful Use, Task Acceleration, Discretionary Time, Holistic Growth.*

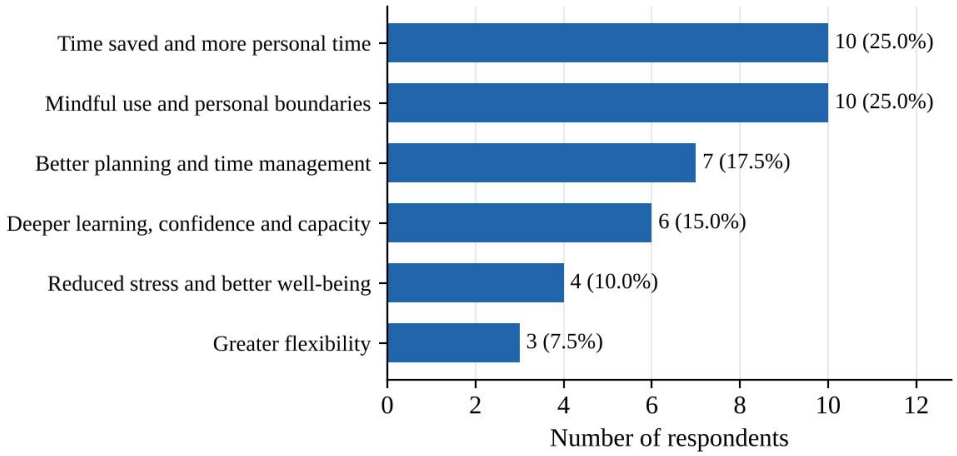


Figure 17: Main themes in responses to Question 17: How AI has changed work–life balance (n = 40)

Decoded Qualitative Synthesis: The 40 open-ended responses to Question 17 provide profound qualitative depth regarding how students experience the daily balance between academic responsibilities and personal life. As detailed in the conceptual coding frame (Table 5, Appendix B) and the 3-level relational coding matrix (Table 6, Appendix B), student reflections coalesce around six core themes. A leading 25.0% of respondents (n = 10) highlighted 'Time saved and more personal time,' describing how delegating routine drafting, syntax checking, and literature searches frees up valuable hours for physical exercise, creative hobbies, and social relationships. One student reflected: 'The time savings from AI have allowed me to pursue additional learning opportunities and hobbies that contribute to my overall personal growth.' Concurrently, an identical 25.0% (n = 10) emphasized 'Mindful use and personal boundaries,' underscoring that balance is not an automatic technological outcome but an intentional discipline. As one respondent articulated: 'While AI helps me be more productive, I've had to learn to set boundaries to prevent work from expanding to fill all available time.'

Other prominent themes include 'Better planning and time management' (17.5%, n = 7), where students utilize AI to build structured study schedules and decompose complex project milestones; 'Deeper learning, confidence and capacity' (15.0%, n = 6), wherein AI provides the cognitive scaffolding required to tackle ambitious research projects without burnout; 'Reduced stress and better well-being' (10.0%, n = 4); and 'Greater flexibility' (7.5%, n = 3). Relational analysis (Table 6) reveals that positive emotional states (Confidence, Relief, Motivation) are triggered when AI acts as an intentional efficiency resource, whereas tension emerges when students realize that without strict personal boundaries, technological speed merely invites additional work.

5.18 Question 18: Qualitative Analysis: Difficulties and Concerns Experienced

Question 18 (Open-Ended): What difficulties or concerns have you experienced while using AI tools in your work or studies?

Keywords: *Over-reliance, Critical Thinking Atrophy, Algorithmic Hallucinations, Verification Fatigue, Academic Integrity Fear, Data Privacy.*

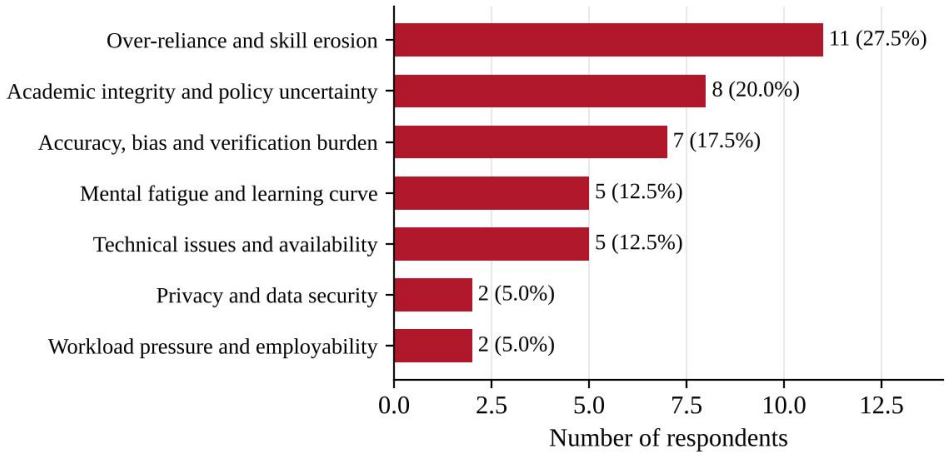


Figure 18: Main themes in responses to Question 18: Difficulties and concerns experienced (n = 40)

Decoded Qualitative Synthesis: Responses to Question 18 reveal the dark underbelly of student AI adoption, clustering into seven rigorous thematic categories (conceptualized in Table 7 and relationally mapped in Table 8, Appendix B). Foremost among student anxieties is 'Over-reliance and skill erosion,' cited by 27.5% of respondents (n = 11). Students expressed profound existential concern regarding the atrophy of their own independent problem-solving and critical thinking faculties. A participant candidly admitted: 'Over-reliance on AI for problem-solving has weakened my independent critical thinking abilities and made me question my own intellectual capacity.' The second most frequent concern was 'Academic integrity and policy uncertainty' (20.0%, n = 8), where students described paralyzing anxiety regarding institutional accusations of plagiarism, false positives from automated AI detectors, and ambiguous faculty guidelines.

A critical cognitive strain was captured by 'Accuracy, bias and verification burden' (17.5%, n = 7) and 'Mental fatigue and learning curve' (12.5%, n = 5). Students highlighted that AI 'hallucinations'—confidently generated falsehoods and fabricated academic citations—force them into an exhausting loop of manual fact-checking. As one respondent noted: 'Hallucination problems where AI confidently presents false information lead to errors in assignments, creating mental fatigue from constantly evaluating output quality.' Technical infrastructure issues (12.5%, n = 5), data privacy risks (5.0%, n = 2), and workload inflation (5.0%, n = 2) constituted additional friction points. Relational analysis indicates that the dominant emotional attitudes are Vulnerability, Paranoia, and Cognitive Fatigue, triggered by algorithmic fallibility and institutional opacity.

5.19 Question 19: Qualitative Analysis: Suggested Institutional Improvements

Question 19 (Open-Ended): What changes would you suggest to your organization or institution so that AI tools support, rather than disrupt, a healthy work-life balance?

Keywords: *Institutional Policy, AI Literacy, Digital Detox, Quiet Zones, Equitable Access, Workload Monitoring, Peer Mentorship.*

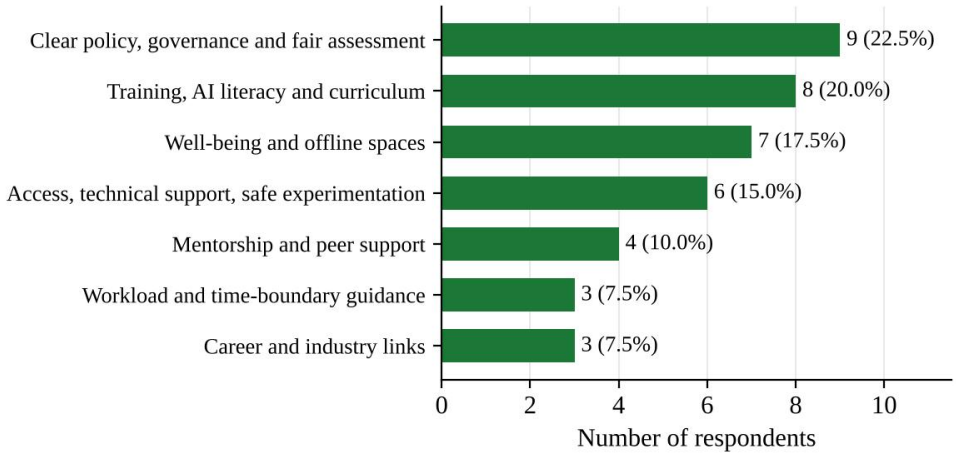


Figure 19: Main themes in responses to Question 19: Suggested institutional improvements (n = 40)

Decoded Qualitative Synthesis: When asked to propose actionable institutional reforms in Question 19, students offered highly pragmatic, multi-dimensional recommendations categorized into seven strategic domains (documented in Table 9 and Table 10, Appendix B). The top priority was 'Clear policy, governance and fair assessment' (22.5%, n = 9), with respondents urging universities to establish unambiguous, regularly updated guidelines defining permitted AI use across different course tasks. One respondent stated: 'Establish clear institutional policies defining acceptable AI use in different contexts, with regular updates as technology evolves and new challenges emerge.' Close behind was 'Training, AI literacy and curriculum' (20.0%, n = 8), demanding mandatory workshops on prompt engineering, information literacy, and critical output evaluation.

Notably, 17.5% of respondents (n = 7) advocated for 'Well-being and offline spaces,' specifically proposing institutional digital wellness initiatives, technology-free quiet study zones, and designated digital detox periods. As one respondent urged: 'Implement regular digital detox periods where students are encouraged to complete assignments without AI assistance to maintain core cognitive skills.' Additional proposals included 'Access, technical support and safe experimentation' (15.0%, n = 6; calling for institutional premium licenses to eliminate economic inequality), 'Mentorship and peer support' (10.0%, n = 4), 'Workload and time-boundary guidance' (7.5%, n = 3; institutional workload caps), and 'Career and industry links' (7.5%, n = 3). Relational analysis demonstrates an underlying attitude of Constructive Advocacy, reflecting a desire to collaborate with university leadership to build a supportive, human-centric academic ecosystem.

5.20 Question 20: Demographic Analysis: Age Group Profile

Question 20: What is your age group? (Single-select categorical: 18–20, 21–23, 24–26, 27–30).

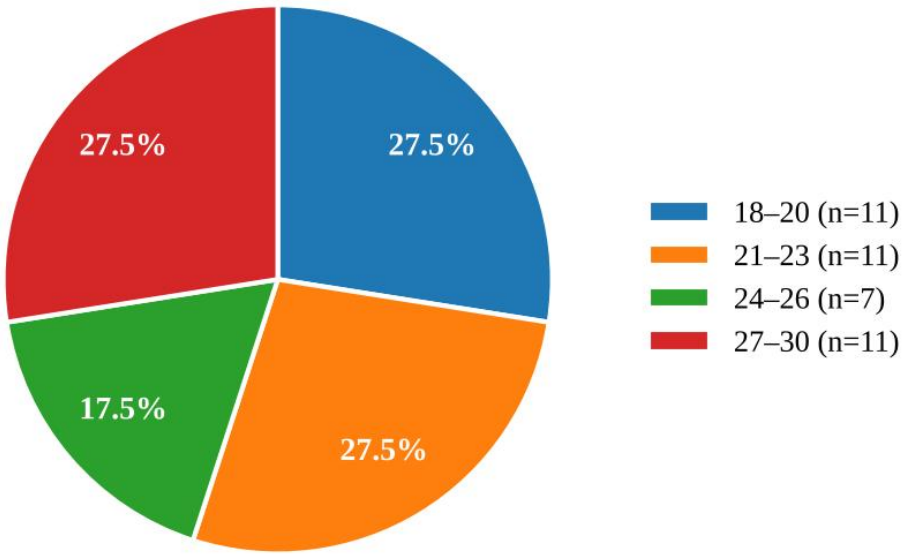


Figure 20: Demographic distribution: Age group of respondents (n = 40)

Quantitative Breakdown and Analytical Significance: The age profile of respondents demonstrates a highly balanced, four-tier demographic spread across the young adult educational spectrum: 27.5% (n = 11) aged 18–20 years, 27.5% (n = 11) aged 21–23 years, 17.5% (n = 7) aged 24–26 years, and 27.5% (n = 11) aged 27–30 years. This distribution ensures that the findings reflect the complete developmental progression of higher education, capturing early-stage undergraduate entrants, graduating seniors, postgraduate master's researchers, and mature scholars balancing advanced education with professional careers.

5.21 Question 21: Demographic Analysis: Current Academic and Professional Status

Question 21: What is your current status? (Single-select categorical).

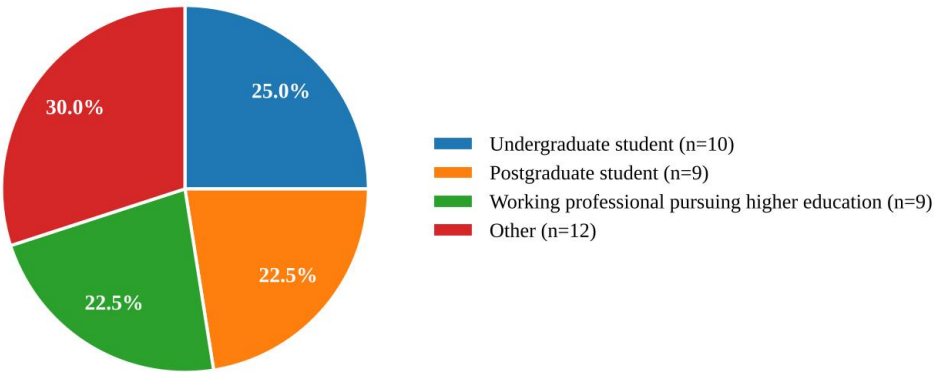


Figure 21: Current academic and professional status of respondents (n = 40)

Quantitative Breakdown and Analytical Significance: Respondents' academic and professional affiliations span all primary tertiary roles: Undergraduate students account for 25.0% (n = 10), Postgraduate students represent 22.5% (n = 9), Working professionals pursuing higher education comprise 22.5% (n = 9), and Research scholars, interns, and dual-role students constitute 30.0% (n = 12). This structural diversity enables the study to evaluate how work–life boundary permeability operates across differing institutional constraints, proving that boundary friction affects full-time students and working scholars alike.

5.22 Question 22: Experience Profile: Duration of Regular AI Tool Use

Question 22: How long have you been using AI tools regularly for work or study? (Single-select categorical).

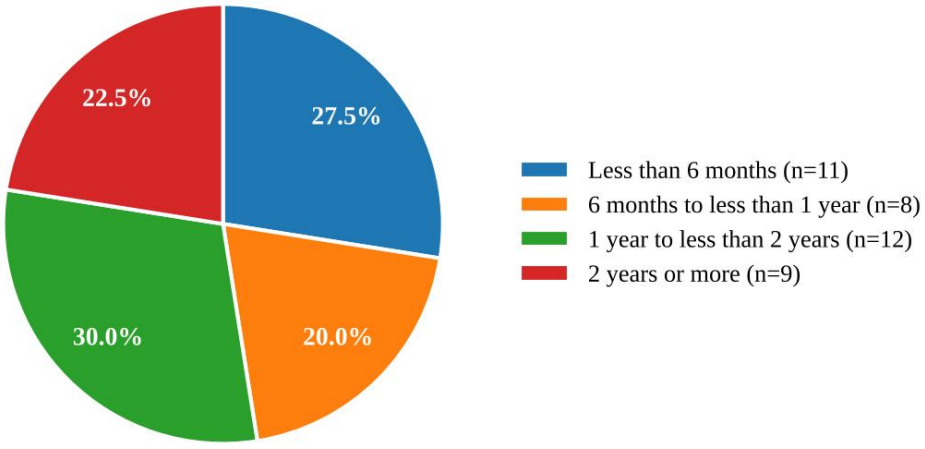


Figure 22: Duration of regular AI tool use among respondents (n = 40)

Quantitative Breakdown and Analytical Significance: Every participant in the sample (100.0%, n = 40) is an active, regular AI user, with zero respondents selecting 'I do not use AI tools.' In terms of adoption tenure, 27.5% (n = 11) have used AI regularly for less

than 6 months, 20.0% (n = 8) for 6 to 12 months, 30.0% (n = 12) for 1 to 2 years, and 22.5% (n = 9) for more than 2 years. Having over half the sample (52.5%) possess more than a year of continuous AI experience ensures that the survey captures mature, entrenched usage habits rather than transient novelty effects.

6. RESULTS SUMMARY

This chapter integrates the empirical findings from Chapter 5 to provide formal, structured answers to the four core Research Questions. Figure 23 visually benchmarks the 14 core quantitative indicators across the four research questions. Following the figure, dedicated narrative syntheses (150–200 words each) address RQ1 through RQ4, concluded by Master Table 4 synthesizing the empirical evidence.

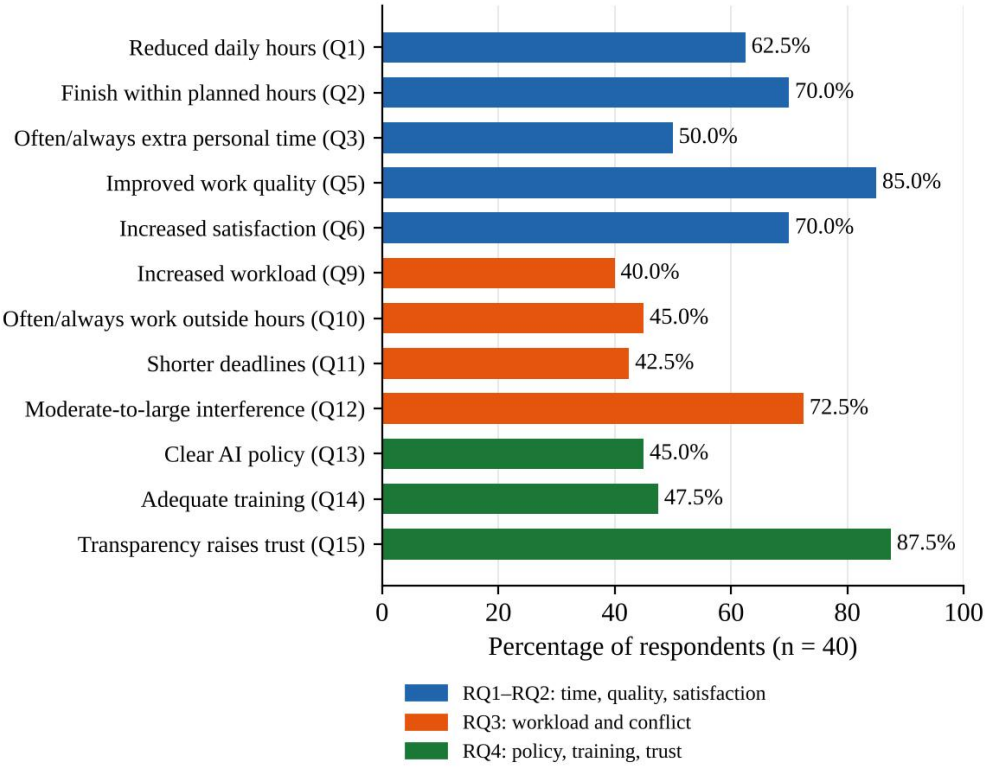


Figure 23: Key survey indicators across the four research questions (percentage of respondents, n = 40)

6.1 Research Question 1: Study-Work Routines and Time Allocation

Research Question 1 Answer (175 words): Empirical evidence demonstrates that Artificial Intelligence significantly reshapes students' study routines and daily task completion, but fails to reliably generate genuine personal leisure. A solid 62.5% of respondents agree that AI reduces total daily study-work hours, and 70.0% report completing commitments within planned hours. However, this efficiency dividend does not translate cleanly into work–life balance: only 50.0% frequently gain extra time for personal activities, while the other half experience personal time rarely or inconsistently. When forced to identify the most noticeable routine change, 35.0% cite flexible working

hours, but 27.5% report longer working hours and 12.5% cite diminished separation between work and life. Crucially, among students who achieve task time reductions, nearly one-third do not gain additional personal time. This confirms that AI accelerates task completion but triggers Jevons' Paradox: surplus time is rapidly absorbed into additional academic tasks, competitive upskilling, and elongated study sessions, leaving personal time constrained.

6.2 Research Question 2: Well-Being, Productivity, and Academic Satisfaction

Research Question 2 Answer (182 words): AI exerts a powerful dual-edged impact on student productivity, output quality, and well-being. On the positive dimension, an overwhelming 85.0% of respondents confirm that AI tools substantially improve the quality of their academic work—the highest performance rating in the study—and 70.0% report heightened overall satisfaction with their academic studies. AI serves as an indispensable cognitive scaffold for brainstorming, literature drafting, and code debugging. Conversely, this productivity dividend incurs heavy psychological and physiological costs. Nearly three-quarters (72.5%) of respondents suffer regular physical discomfort, such as eye strain and tension headaches, stemming from prolonged, vigilant screen engagement during AI verification. Alarmingly, when identifying the single most significant effect of AI, 40.0% of respondents cite intellectual vulnerability—specifically, 22.5% report heightened anxiety regarding the future value of their skills, and 17.5% experience diminished confidence in their independent problem-solving capacity. Thus, while AI elevates external output quality, it simultaneously threatens physical health and induces cognitive dependency.

6.3 Research Question 3: Workload Intensification and Boundary Permeability

Research Question 3 Answer (180 words): Artificial Intelligence contributes substantially and systematically to academic workload intensification, temporal boundary erosion, and work–life conflict. A full 40.0% of respondents report that expected work volumes have expanded since AI adoption, and 42.5% confirm that assignment deadlines have become noticeably shorter, reflecting institutional expectations that technological tools warrant accelerated delivery. Most critically, academic boundaries have severely deteriorated: 45.0% of students work outside normal study hours on a regular basis (often or always), with an additional 40.0% doing so occasionally, leaving only 2.5% who never work after hours. Consequently, 72.5% of respondents experience moderate to large interference with their personal and family life. The constant, ubiquitous availability of AI on personal smartphones creates an invasive tether to academic obligations, transforming bedrooms and rest periods into continuous work zones. AI efficiency does not alleviate pressure; rather, it facilitates round-the-clock academic engagement that directly disrupts domestic life and personal relationships.

6.4 Research Question 4: Institutional Policies, Training, and Ethical Safeguards

Research Question 4 Answer (178 words): The investigation reveals that institutional governance frameworks and pedagogical support have lagged far behind rapid student AI adoption, generating acute ethical friction. Fewer than half of respondents (45.0%) report

that their university possesses a clear policy on acceptable AI use, and only 47.5% have received formal training on responsible AI practices. This institutional ambiguity leaves students navigating academic integrity risks independently, fearing automated plagiarism accusations. In response, an emphatic 87.5% of respondents assert that transparent, ethical AI practices would substantially increase their institutional trust. When prioritizing organizational interventions, 32.5% demand structural time and workload protections (such as explicit right-to-disconnect rules and realistic deadlines accounting for verification), 22.5% seek formal information literacy and AI ethics training, and 20.0% advocate for well-being initiatives including digital detox periods and AI-free quiet zones. Maintaining a sustainable work–life balance requires universities to move beyond punitive honor codes and implement structured, human-centred AI governance.

Table 4: Master summary of key survey indicators across the four research questions (n = 40)

Research Question	Core Survey Indicator	Empirical Survey Result (n = 40)
RQ1: Routines & Time	Reduced total daily work or study hours (Q1)	62.5% Agree (47.5% agree, 15.0% strongly agree)
RQ1: Routines & Time	Finished commitments within planned hours (Q2)	70.0% Agree (50.0% agree, 20.0% strongly agree)
RQ1: Routines & Time	Extra time for personal activities often or always (Q3)	50.0% Positive (37.5% often, 12.5% always)
RQ1: Routines & Time	Longer hours or less boundary separation (Q4)	40.0% Combined (27.5% longer hours, 12.5% less separation)
RQ2: Well-Being & Output	Improved the quality of work produced (Q5)	85.0% Agree (57.5% agree, 27.5% strongly agree)
RQ2: Well-Being & Output	Increased overall satisfaction with studies (Q6)	70.0% Agree (47.5% agree, 22.5% strongly agree)
RQ2: Well-Being & Output	Physical discomfort experienced at least sometimes (Q7)	72.5% Report Discomfort (55.0% sometimes, 17.5% often/always)
RQ2: Well-Being & Output	Skill-related worry or reduced confidence as main effect (Q8)	40.0% Insecurity (22.5% skill worry, 17.5% confidence loss)
RQ3: Workload & Conflict	Expected volume of work has increased (Q9)	40.0% Agree (32.5% agree, 7.5% strongly agree)
RQ3: Workload & Conflict	Work outside normal hours often or always (Q10)	45.0% High Frequency (35.0% often, 10.0% always)
RQ3: Workload & Conflict	Deadlines have become shorter (Q11)	42.5% Agree (30.0% agree, 12.5% strongly agree)
RQ3: Workload & Conflict	Moderate to large interference with personal/family life (Q12)	72.5% Interference (57.5% moderate, 15.0% large extent)
RQ4: Institutional Strategy	Institution has a clear policy on acceptable AI use (Q13)	45.0% Agree (Deficit: 55.0% neutral or disagree)
RQ4: Institutional Strategy	Transparent and ethical AI practices would increase trust (Q15)	87.5% Agree (45.0% agree, 42.5% strongly agree)

7. CONCLUSION

7.1 Synthesis of Findings

This empirical technical report establishes that Artificial Intelligence exerts a powerful, highly contradictory influence on the work–life balance of higher education students aged 18 to 30. AI is neither an unmitigated disaster nor an unalloyed blessing; rather, it operates as a classic double-edged socio-technical phenomenon. On one side, students achieve undeniable cognitive acceleration: 85.0% produce superior academic work, 70.0% adhere to planned completion schedules, and 62.5% experience task-level time savings. On the other side, these efficiency dividends rarely translate into sustainable personal leisure. Due to Jevons' Paradox and unbounded connectivity, 72.5% of students suffer moderate-to-large interference with family and domestic life, 45.0% routinely study after hours, 72.5% experience physical discomfort, and 40.0% grapple with intellectual insecurity and cognitive dependency. AI accelerates academic output, but simultaneously expands workload expectations and demolishes temporal boundaries.

7.2 Pedagogical and Institutional Implications

The empirical insights carry profound policy implications for universities and tertiary educational leadership. First, institutions must urgently transition away from ambiguous, punitive AI policies toward transparent, task-specific governance frameworks. Clear institutional demarcation must define where AI is encouraged (conceptual brainstorming, formatting, code optimization) versus where independent cognitive synthesis is required. Second, universities must introduce explicit 'Right-to-Disconnect' institutional protocols. Faculty and project supervisors must not schedule assignment deadlines during late-night hours or weekend mornings, nor should they compress deadlines under the assumption that AI automates cognitive work. Third, academic curricula must integrate formal AI literacy training, teaching students not only advanced prompt architecture but also rigorous hallucination verification techniques, digital wellness practices, and strategies to guard against intellectual atrophy.

7.3 Recommendations for Higher Education Students

For students navigating this demanding environment, the report provides three essential recommendations. First, students must practice intentional boundary setting. Discretionary time generated through AI efficiency must be actively ring-fenced for physical exercise, restorative sleep, and social relationships, rather than being passively absorbed into additional coursework. Second, students must resist passive algorithmic outsourcing. To preserve problem-solving resilience, students should struggle independently with difficult conceptual challenges before prompting AI systems, maintaining cognitive ownership over their intellectual development. Third, students should adopt strict digital hygiene routines, enforcing screen breaks to mitigate visual strain and designating AI-free study intervals to foster deep contemplative learning.

7.4 Study Limitations and Directions for Future Longitudinal Research

While this study delivers rigorous mixed-methods insights, several avenues for future research remain. Future inquiries should deploy longitudinal cohort designs to track how student work–life balance, GPA metrics, and cognitive problem-solving resilience evolve over multi-year degree programs under varying AI conditions. Furthermore, comparative cross-institutional studies across medical, legal, humanities, and engineering faculties will elucidate discipline-specific boundary dynamics. Investigating the efficacy of institutional pilot interventions—such as mandatory AI-free detox weeks and institutional workload caps—will provide vital evidence to ensure that Artificial Intelligence empowers, rather than disrupts, the flourishing of future scholars.

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APPENDIX A – QUESTIONNAIRE

Survey Topic: Is Artificial Intelligence Disrupting the Work–Life Balance of Higher Education Students (Age 18–30)?
Target Population: Higher education students aged 18–30 who utilize AI-enabled tools in academic coursework, research, internships, or employment.
Administration: Online Google Forms questionnaire; participation is completely voluntary and responses are anonymous.

Section A: Work–Life Balance and AI Adoption (RQ1)

1. Using AI tools has reduced the total number of hours I spend on my daily work or study tasks.
Options: [ ] Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree

2. Since adopting AI tools, I am able to finish my work or study commitments within my planned hours.
Options: [ ] Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree

3. How often do you have extra time for personal activities (e.g., family, hobbies, rest, exercise) since you started using AI tools?
Options: [ ] Never [ ] Rarely [ ] Sometimes [ ] Often [ ] Always

4. Which change in your daily routine has been most noticeable since you started using AI tools? (Select one)
Options: [ ] Shorter working or study hours [ ] Longer working or study hours [ ] More flexibility in when I work [ ] Less separation between work time and personal time [ ] No noticeable change

Section B: Well-Being, Productivity and Satisfaction (RQ2)

5. AI-enabled tools have improved the quality of the work I produce.
Options: [ ] Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree

6. Using AI tools has increased my overall satisfaction with my work or studies.
Options: [ ] Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree

7. How often do you experience physical discomfort (e.g., eye strain, headaches) while using AI tools?
Options: [ ] Never [ ] Rarely [ ] Sometimes [ ] Often [ ] Always

8. Which effect of AI-enabled tools is the most significant in your own experience? (Select one)
Options: [ ] Higher productivity [ ] Better quality of output [ ] Increased worry about the value of my own skills [ ] Reduced confidence in my independent problem-solving [ ] Greater interest and engagement in my work [ ] No significant effect

Section C: Workload and Work–Life Conflict (RQ3)

9. Since AI tools were introduced, the volume of work expected from me has increased.
Options: [ ] Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree

10. How often do you work on tasks outside your normal working or study hours?
Options: [ ] Never [ ] Rarely [ ] Sometimes [ ] Often [ ] Always

11. Since AI tools were introduced, the deadlines given to me for my work or study tasks have become shorter.
Options: [ ] Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree

12. To what extent do AI-related work or study demands interfere with your personal and family life?

Options: [ ] Not at all [ ] To a small extent [ ] To a moderate extent [ ] To a large extent [ ] To a very large extent

Section D: Organisational Strategies and Ethical AI Practices (RQ4)

13. My organization or institution has a clear policy on the acceptable use of AI tools.

Options: [ ] Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree

14. I have received adequate training or guidance on using AI tools responsibly.

Options: [ ] Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree

15. Transparent and ethical AI practices, such as clear disclosure of how AI decisions are made, would increase my trust in AI tools at work or in study.

Options: [ ] Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree

16. Which organizational measure would most help you maintain a healthy work-life balance while using AI tools? (Select one)

Options: [ ] A clear right-to-disconnect policy outside working hours [ ] Realistic deadlines that account for verifying AI output [ ] Formal training on effective and ethical AI use [ ] Transparency about how AI is used to monitor or evaluate performance [ ] Flexible working or study hours [ ] None of the above / Other institutional measure

Section E: Qualitative Open-Ended Questions

17. In your own words, how has using Artificial Intelligence tools changed the way you balance your work or studies with your personal life?

Response format: Open-ended written narrative response.

18. What difficulties or concerns have you experienced while using AI tools in your work or studies?

Response format: Open-ended written narrative response.

19. What changes would you suggest to your organization or institution so that AI tools support, rather than disrupt, a healthy work-life balance?

Response format: Open-ended written narrative response.

Section F: Demographic and Background Information

20. What is your age group?

Options: [ ] 18–20 [ ] 21–23 [ ] 24–26 [ ] 27–30

21. What is your current status?

Options: [ ] Undergraduate student [ ] Postgraduate student [ ] Research scholar [ ] Student with an internship or part-time job [ ] Working professional pursuing higher education [ ] Other

22. How long have you been using AI tools regularly for work or study?

Options: [ ] I do not use AI tools [ ] Less than 6 months [ ] 6 months to less than 1 year [ ] 1 year to less than 2 years [ ] 2 years or more

APPENDIX B – SUPPORTING TABLES

This appendix compiles the complete suite of qualitative content analysis coding frames and 3-level relational coding matrices developed to evaluate the open-ended responses for Questions 17, 18, and 19. Conceptual coding establishes thematic frequencies, while relational coding maps emotional attitudes (Level 1), technological/institutional triggers (Level 2), and underlying socio-pedagogical meanings (Level 3).

Table 5: Conceptual analysis: coding frame for Question 17 (Changes in work–life balance, n = 40)

Thematic Category / Code	Description of What Respondents Described	Frequency (n, %)
Time saved and more personal time	Faster completion of routine tasks leaving surplus time for hobbies, exercise, family, and self-care	10 (25.0%)
Mindful use and personal boundaries	Deliberate boundary discipline, avoiding intellectual dependency, and preserving personal time	10 (25.0%)
Better planning and time management	Structured study schedules, systematic project decomposition, and superior priority setting	7 (17.5%)
Deeper learning, confidence and capacity	More bandwidth for advanced critical analysis; confidence to pursue challenging milestones	6 (15.0%)
Reduced stress and better well-being	Mitigation of time pressure, exam-period anxiety reduction, and improved mental well-being	4 (10.0%)
Greater flexibility	Freedom to dynamically align academic work with personal energy rhythms and internships	3 (7.5%)

Table 6: Relational analysis: 3-level thematic coding matrix for Question 17 (Changes in balance)

Representative Response (Verbatim)	Level 1: Emotion / Attitude	Level 2: Trigger / Catalyst	Level 3: Underlying Meaning
"The time savings from AI have allowed me to pursue additional learning opportunities and hobbies that contribute to my overall personal growth."	**Relief and Empowerment**	Fast automation of routine drafting and literature search	AI functions as a transformative resource when saved hours are actively converted into personal development.
"While AI helps me be more productive, I've had to learn to set boundaries to prevent work from expanding to fill all available time."	**Vigilance and Discipline**	Risk of continuous work expansion	Work–life balance requires deliberate cognitive boundaries; technological efficiency alone naturally expands work.
"AI assistance has helped me create more structured study schedules, which has improved both my academic performance and personal time management."	**Sense of Control**	Algorithmic scheduling and task decomposition	Predictability in study workflows eliminates ambient academic anxiety, safeguarding domestic personal time.
"AI tools have helped me develop better stress management strategies by reducing the time pressure associated with routine academic tasks."	**Calm and Reassurance**	Reduction of deadline time pressure	Alleviating operational friction prevents academic panic from spilling over into sleep hygiene and domestic life.
"The support of AI has given me confidence to tackle more challenging academic goals while maintaining a	**Self-Efficacy and Ambition**	Instant cognitive scaffolding and feedback	AI enhances academic capability without demanding unsustainable sacrifices in

sustainable lifestyle."	personal health and well-being.

Table 7: Conceptual analysis: coding frame for Question 18 (Difficulties and concerns experienced, n = 40)

Thematic Category / Code	Description of What Respondents Described	Frequency (n, %)
Over-reliance and skill erosion	Weakened critical thinking, memory retention, creativity, and fear of intellectual dependency	11 (27.5%)
Academic integrity and policy uncertainty	Ambiguous institutional rules, fear of false plagiarism flags, and citation attribution confusion	8 (20.0%)
Accuracy, bias and verification burden	Algorithmic hallucinations, fabricated citations, bias, and the exhausting burden of fact-checking	7 (17.5%)
Mental fatigue and learning curve	Cognitive exhaustion from continuous prompt engineering, output evaluation, and task-switching	5 (12.5%)
Technical issues and availability	Token limits, API throttling, server downtime, and connectivity disruptions during critical tasks	5 (12.5%)
Privacy and data security	Unease regarding the uploading of proprietary research, code, or personal data to third-party clouds	2 (5.0%)
Workload pressure and employability	Efficiency paradoxically creating pressure for higher output and anxiety over future skill relevance	2 (5.0%)

Table 8: Relational analysis: 3-level thematic coding matrix for Question 18 (Difficulties and concerns)

Representative Response (Verbatim)	Level 1: Emotion / Attitude	Level 2: Trigger / Catalyst	Level 3: Underlying Meaning
"Over-reliance on AI for problem-solving has weakened my independent critical thinking abilities and made me question my own intellectual capacity."	**Self-Doubt and Imposter Anxiety**	Habitual outsourcing of core intellectual problem-solving	Convenience creates cognitive dependency, degrading autonomous self-confidence and academic resilience.
"Workplace and university policy uncertainty creates fear of violating institutional guidelines about AI use in academic contexts."	**Apprehension and Vulnerability**	Ambiguous institutional integrity definitions and punitive honor codes	Absence of clear guidelines transforms everyday study into an ethically risky, high-anxiety endeavor.
"Hallucination problems where AI confidently presents false information lead to errors in research projects, requiring constant verification."	**Exhaustion and Skepticism**	Algorithmic fabrications and confidently incorrect citations	Verification overhead offsets speed gains; cognitive labor shifts from generation to exhausting error-policing.
"Mental fatigue from constantly evaluating AI output quality and making decisions about when to trust or modify generated content."	**Cognitive Strain**	Continuous judgment calls on algorithmic trustworthiness	High-velocity token generation forces sustained near-distance cognitive monitoring, depleting mental reserves.
"Time management issues where AI efficiency creates pressure to take on more work, ultimately leading to increased stress rather than time savings."	**Frustration and Entrapment**	Jevons' Paradox: increased capacity driving heightened task volume	Technological acceleration ratchets up baseline productivity norms, eliminating true rest.

Table 9: Conceptual analysis: coding frame for Question 19 (Suggested institutional changes, n = 40)

Thematic Category / Code	Description of What Respondents Proposed	Frequency (n, %)
Clear policy, governance and fair assessment	Comprehensive acceptable-use frameworks, transparent evaluation protocols, and flexible assessment rubrics	9 (22.5%)
Training, AI literacy and curriculum	Workshops on information literacy, prompt engineering, bias detection, and ethical usage courses	8 (20.0%)
Well-being and offline spaces	Institutional digital wellness workshops, stress management support, quiet study zones, and digital detox days	7 (17.5%)
Access, technical support and safe experimentation	Institutional premium licenses to bridge economic divides, sandbox environments, and IT helpdesks	6 (15.0%)
Mentorship and peer support	Peer mentorship networks and collaborative discussion forums on maintaining healthy technology boundaries	4 (10.0%)
Workload and time-boundary guidance	Institutional workload monitoring caps and guidelines preventing late-night electronic submissions	3 (7.5%)
Career and industry links	Career counseling demonstrating how human skills complement AI, along with industry partnership insights	3 (7.5%)

Table 10: Relational analysis: 3-level thematic coding matrix for Question 19 (Suggested institutional changes)

Representative Response (Verbatim)	Level 1: Emotion / Attitude	Level 2: Trigger / Catalyst	Level 3: Underlying Meaning
"Establish clear institutional policies defining acceptable AI use in different contexts, with regular updates as technology evolves."	**Constructive Advocacy**	Absence of transparent academic guidelines	Proactive administrative governance restores psychological safety, removing fear of unfair integrity penalties.
"Provide workshops on information literacy that help students evaluate AI-generated content for accuracy, bias, and reliability."	**Desire for Mastery**	Exhaustion from hallucination verification	Structured pedagogical training builds critical verification competence, converting trial-and-error into structured skill.
"Implement regular digital detox periods where students are encouraged to complete assignments without AI assistance to maintain core skills."	**Longing for Cognitive Balance**	Uninterrupted digital connectivity and skill anxiety	Institutional mandates for unplugged study safeguard fundamental human problem-solving and emotional health.
"Provide access to premium AI tools through institutional licenses to ensure equitable access and reduce financial stress."	**Sense of Justice**	Economic inequality in accessing advanced LLM models	Equitable institutional access eliminates financial barriers, preventing academic divides between wealthy and under-resourced peers.
"Implement workload monitoring systems to ensure AI efficiency gains don't lead to unrealistic expectations and increased student stress levels."	**Protection and Boundary Defense**	Escalating baseline assignment expectations	Administrative workload caps are vital to ensure technology serves human well-being rather than unbridled academic hyper-production.